THE FULL ADJOINT-GENERATED CASE OF GINIBRE'S CONDITION Q_3 PART III: COMPLETION OF THE FULL SIGNED-PRODUCT HIERARCHY

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ABSTRACT. Let G be a compact connected Lie group with simple Lie algebra and let $\mathcal{Q}_G$ be the closed multiplicative convex cone generated by its real adjoint character. We reduce Ginibre's full condition Q_3 on $\mathcal{Q}_G$ to one explicit two-parameter hierarchy and prove that hierarchy for every compact connected G with simple Lie algebra. The proof treats every finite tuple length, every sign assignment, and every even number of minus factors; Ginibre's Proposition 1 then promotes the adjoint generator to $\mathcal{Q}_G$. The last finite classical boxes are closed by bounded Littlewood determinant identities and an exact prime-field/GMP certificate covering 17,862 residual inequalities. Here "full" refers to Ginibre's finite-tuple and sign-pattern quantifiers on the explicitly stated adjoint-generated cone; it does not mean the cone of all positive-definite functions. We also give an exact SO(3) counterexample showing that Ginibre's abelian cone of all real positive-definite functions cannot be carried over literally to nonabelian compact groups.

Remark 0.1 (Submission architecture and audit status)**.** The formal proof has three numbered parts and one detailed supplement collected in three files: `paper.pdf` consolidates Parts I–II, `paper_full.pdf` is their active computational supplement, and the present file is Part III. All three files are load-bearing. In particular, the detailed supplement supplies the numbered B/C correction-prefix proofs and half-stable bridge reduction used by the compact residual-closure proposition in `paper.pdf`. The concise dependency spine is recorded in `PUBLICATION_PROOF_SPINE.md`. Diagnostic and superseded routes are not printed in the formal supplement and are not inputs to that spine; the source-only development archive can be typeset separately by defining `\GinibreDevelopmentArchive`. Files under `referee_audits/` and the historical report in the source archive are author-controlled self-audits. They are reproducibility aids, not independent peer review. The theorem depends only on the numbered results and authenticated computations mapped in the code and data availability section.

1. The exact adjoint-generated target

The cone studied here is the smallest uniformly closed cone containing the adjoint character and closed under multiplication and nonnegative linear combination. It is natural for two independent reasons. Representation theoretically, its polynomial part consists exactly of nonnegative linear combinations of tensor-power characters $1, \chi_{\mathrm{ad}}, \chi_{\mathrm{ad}}^2, \ldots$; hence its Haar moments are invariant-tensor multiplicities in tensor powers of the adjoint module. From the viewpoint of Ginibre's Proposition 1, it is precisely the generated cone to which the single-generator proof promotes. A statement on any larger cone requires additional inequalities not supplied by that promotion. The theorem therefore isolates the full signed-product consequences of the adjoint tensor category. It does not assert that a general central positive-definite function is a uniform limit of these polynomials, and it supplies no conclusion for a model whose interaction or observables do not belong to this cone.

Let G be as in the abstract, let μ_G be normalized Haar measure, and write $X = \chi_{\mathrm{ad}}(g)$ and $Y = \chi_{\mathrm{ad}}(h)$ for independent Haar variables. Since the adjoint representation is real, X and Y are real-valued.

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Definition 1.1 (Adjoint-generated cone)**.** The adjoint-generated multiplicative convex cone is

$$\mathcal{Q}_G = \overline{\left\{ \sum_{j=0}^{d} c_j \chi_{\mathrm{ad}}^j : d \geq 0,\ c_j \geq 0 \right\}}^{\|\cdot\|_\infty} \subset C(G).$$

For integers $a, n \geq 0$, define

$$I_{a,n}^G = \int_{G\times G} (\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^{2a} (\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n \, d\mu_G(g) d\mu_G(h).$$

Lemma 1.2 (The adjoint-generated cone is positive definite)**.** *Every element of $\mathcal{Q}_G$ is a real continuous positive-definite function on G. In particular, $\mathcal{Q}_G$ is a subcone of the cone $\mathcal{P}_G$ used in Proposition 1.5 below.*

Proof. Choose an orthonormal basis (e_j) of the real adjoint module. Then

$$\chi_{\mathrm{ad}}(g) = \sum_j \langle e_j, \mathrm{Ad}(g) e_j \rangle$$

is positive definite: for arbitrary $g_1, \dots, g_r \in G$ and $z_1, \dots, z_r \in \mathbb{C}$ its defining quadratic form is

$$\sum_j \left\| \sum_{k=1}^{r} z_k \,\mathrm{Ad}(g_k) e_j \right\|^2 \geq 0.$$

Products of positive-definite functions are positive definite by tensoring their unitary representations, and nonnegative sums are positive definite by direct sums and scaling. Finally, a uniform limit preserves every finite positive-semidefinite Gram matrix entrywise. The character and all its powers are real and continuous, proving the assertion from the definition of $\mathcal{Q}_G$. □

Theorem 1.3 (Full Q_3 reduction on the adjoint-generated cone)**.** *Ginibre's condition Q_3 holds for $\mathcal{Q}_G$ if and only if*

$$I_{a,n}^G \geq 0 \qquad (a, n \geq 0).$$

The only nonautomatic members of this hierarchy are

$$a \geq 1, \qquad n \text{ odd}.$$

Theorem 1.4 below is exactly the row $a = 1$.

Proof. For the singleton generating set $S_G = \{\chi_{\mathrm{ad}}\}$, a signed product with r minus factors and n plus factors is

$$\int_{G\times G} (X - Y)^r (X + Y)^n \, d\mu_G \, d\mu_G.$$

It vanishes when r is odd, by interchanging X and Y, and for $r = 2a$ it is $I_{a,n}^G$. Thus the displayed hierarchy is precisely condition Q_3 for the generating set S_G.

For completeness, the passage from the generator to the cone uses the factorization

$$\begin{aligned}(fg)(x) \pm (fg)(y) = \frac{1}{2}\big[&(f(x) + f(y))(g(x) \pm g(y)) \\ &+ (f(x) - f(y))(g(x) \mp g(y))\big].\end{aligned}$$

Iterating this identity expands every signed factor belonging to a product of generators as a sum, with nonnegative coefficients, of products of signed generator factors. Convex combinations preserve the inequality, and uniform limits preserve it because Haar measure is finite. This is the argument of Ginibre [3, Proposition 1 and (1.5)]; it proves that S_G satisfies Q_3 if and only if $\mathcal{Q}_G$ does.

It remains to identify the automatic sectors. If n is even, the integrand defining $I^G_{a,n}$ is pointwise nonnegative. When $a = 0$, expand

$$\mathbf{E}(X+Y)^n = \sum_{j=0}^{n} \binom{n}{j} m_j^G m_{n-j}^G, \qquad m_j^G = \dim\bigl((\mathfrak{g}_{\mathbb{C}}^{\otimes j})^G\bigr) \geq 0.$$

Hence the all-plus sector is nonnegative as well. □

Theorem 1.4 (Two-minus adjoint-character theorem)**.** *For every compact connected Lie group G with simple Lie algebra and every integer $n \geq 0$,*

$$I^G_{1,n} \geq 0.$$

Proof. This is the main theorem of the journal-facing Parts I–II manuscript, [1, Theorem 2.2]. In the notation of that theorem its functional $Q_3^G(n)$ is exactly $I^G_{1,n}$ above. That manuscript is generated as `paper.pdf` from `paper.tex` and is submitted with the present Part III. The detailed computational supplement built from the same source supplies the numbered correction-prefix proofs and the half-stable bridge reduction used by the compact manuscript. □

Proposition 1.5 (The all-positive-definite analogue is false)**.** *Let $G = \mathrm{SO}(3)$ with normalized Haar measure, and let $\mathcal{P}_G$ denote the cone of all real continuous positive-definite functions on G. Then $\mathcal{P}_G$ does not satisfy Ginibre's condition Q_3. More precisely, four functions in $\mathcal{P}_G$ and the sign pattern $--++$ give the value*

$$-\frac{8}{279}.$$

Consequently the adjoint-generated restriction in Theorem 1.3 cannot be removed, even for a compact connected group with simple Lie algebra.

Proof. For a unit vector $v \in \mathbb{R}^3$, put

$$f_v(g) = \langle v, gv \rangle, \qquad g \in \mathrm{SO}(3).$$

This function is real and continuous. It is positive definite because, for arbitrary $g_1, \ldots, g_r \in G$ and $z_1, \ldots, z_r \in \mathbb{C}$,

$$\sum_{i,j=1}^{r} \overline{z_i} z_j f_v(g_i^{-1} g_j) = \left\| \sum_{j=1}^{r} z_j g_j v \right\|^2 \geq 0.$$

We first record the Haar integrals needed below. For unit vectors $v_1, \ldots, v_4$, define

$$c_1 = \langle v_1, v_2 \rangle \langle v_3, v_4 \rangle, \quad c_2 = \langle v_1, v_3 \rangle \langle v_2, v_4 \rangle, \quad c_3 = \langle v_1, v_4 \rangle \langle v_2, v_3 \rangle.$$

Haar projection on the standard representation gives

$$\int_G f_{v_i} = 0, \qquad \int_G f_{v_i} f_{v_j} = \frac{\langle v_i, v_j \rangle^2}{3}.$$

For the fourth integral, the three pairing tensors in $(\mathbb{R}^3)^{\otimes 4}$ have Gram matrix with diagonal entries 9 and off-diagonal entries 3. Its inverse has diagonal entries $2/15$ and off-diagonal entries $-1/30$. Since Haar averaging is the orthogonal projection onto the invariant tensors, contraction with $v_1 \otimes \cdots \otimes v_4$ on both sides gives

$$\int_G \prod_{i=1}^{4} f_{v_i} = \frac{2}{15} \sum_{i=1}^{3} c_i^2 - \frac{1}{15} \sum_{1 \leq i < j \leq 3} c_i c_j. \tag{1}$$

Take

$$v_1 = \left(\sqrt{\frac{19}{31}}, \sqrt{\frac{12}{31}}, 0 \right), \quad v_2 = \left(\sqrt{\frac{19}{31}}, -\sqrt{\frac{12}{31}}, 0 \right), \quad v_3 = v_4 = (1,0,0).$$

Then $(c_1, c_2, c_3) = (7, 19, 19)/31$, and (1) equals $61/961$. Expanding the signed product and using the vanishing one-point integrals yields, with $M_{ij} = \int_G f_{v_i} f_{v_j}\, d\mu_G$,

$$\begin{aligned}
\mathcal{J} = \int_{G\times G} &(f_{v_1}(g) - f_{v_1}(h)) \\
&\cdot (f_{v_2}(g) - f_{v_2}(h)) \\
&\cdot (f_{v_3}(g) + f_{v_3}(h)) \\
&\cdot (f_{v_4}(g) + f_{v_4}(h)) \\
= 2 &\left[\int_G \prod_{i=1}^{4} f_{v_i}\, d\mu_G + M_{12}M_{34} - M_{13}M_{24} - M_{14}M_{23}\right] \\
= 2 &\left[\frac{61}{961} + \frac{49 - 361 - 361}{9 \cdot 961}\right] = -\frac{8}{279}.
\end{aligned}$$

This is a strict violation of Q_3 inside $\mathcal{P}_G$. □

Remark 1.6 (Comparison with Ginibre's abelian Example 4)**.** Ginibre [3, Example 4] proves Q_3 for the cone of all real positive-definite functions when the compact group is a finite product of circles and finite cyclic groups. At the end of that paper he proposes the same cone on a noncommutative compact group only as a candidate. Proposition 1.5 shows that this proposed extension is false. The result proved here matches Ginibre's complete Q_3 quantifiers and his Proposition 1 cone-promotion mechanism, but on the explicitly smaller multiplicative cone generated by the adjoint character. The matrix coefficients in Proposition 1.5 are not central. Consequently neither that counterexample nor the affirmative theorem decides Q_3 for the intermediate cone of all real continuous central positive-definite functions on a nonabelian compact group.

Remark 1.7 (Relation to earlier nonabelian correlation inequalities)**.** Sylvester [4, Secs. 1, 3–4] had already disproved the general Ginibre inequality for higher-dimensional $O(N)$ spin systems by constructing graph-cycle counterexamples. His variables are spins on a product of spheres and his signed factors are indexed by graph edges. The present Proposition 1.5 supplies a different, exact witness directly in the compact-group formulation proposed at the end of Ginibre's 1970 paper: four explicit real positive-definite functions on SO(3), integrated over two normalized Haar variables, give $-8/279$. We therefore make no claim that this is the first nonabelian failure of a Ginibre inequality.

Abdesselam [5, Sec. 1 and Thms. 2.1–2.3] subsequently formulated general and padded Ginibre inequalities for invariant observables of $O(N)$ and $\mathbb{CP}^{N-1}$ models. He proves large-power asymptotics in terms of Kirchhoff polynomials and proves all padded Ginibre inequalities for inverse half-integer powers of stable determinantal polynomials. Those results concern an asymptotic regime and a determinantal-polynomial substitute; they do not establish the finite-rank compact-group Haar hierarchy proved here. Conversely, the present theorem is restricted to the adjoint-generated cone and does not settle the invariant-observable GKS2 problem for finite $O(N)$ spin systems, which Abdesselam records as open for $N \geq 3$.

Remark 1.8 (Scope of potential quantum applications)**.** Theorem 8.22 is an algebraic Haar-positivity theorem. Through Ginibre's general mechanism it can serve as an input for group-valued lattice models whose interactions genuinely lie in $\mathcal{Q}_G$. It does not by itself prove correlation inequalities for quantum Heisenberg or general $O(N)$ vector-spin systems, nonlinear sigma models, lattice gauge theories, or continuum quantum field theories. In particular, the occurrence of an adjoint representation in a gauge model neither places its plaquette interactions and observables in $\mathcal{Q}_G$ nor removes the shared-link constraints. This distinction is visible in the recent work of Chevyrev and Garban [6, Thm. 1.5, Cor. 1.6, and Rem. 1.7]: their Villain-action limit applies to nonabelian lattice gauge theories, while the Ginibre monotonicity consequence established there is abelian.

Such applications require additional model-specific results: Gibbs-measure promotion on the relevant observable cone, a thermodynamic-limit argument, and, for a quantum reconstruction, a reflection-positive transfer-matrix or equivalent construction. A direct route toward the conventional finite-N $O(N)$ problem would instead require the two-minus or padded inequality for defining-representation matrix coefficients on products of spheres. That is a separate theorem and is not asserted here.

Lemma 1.9 (Moment and generating-function forms). *For $a, n \geq 0$,*

$$I^G_{a,n} = \sum_{j=0}^{2a} \sum_{k=0}^{n} (-1)^j \binom{2a}{j} \binom{n}{k} m^G_{2a-j+k} m^G_{j+n-k}.$$

If

$$M_G(t) = \mathbf{E} e^{tX} = \sum_{r \geq 0} m^G_r \frac{t^r}{r!},$$

then, as an identity of convergent power series,

$$F^G_a(t) := \mathbf{E}\big[(X-Y)^{2a} e^{t(X+Y)}\big] = \sum_{j=0}^{2a} (-1)^j \binom{2a}{j} M_G^{(2a-j)}(t) M_G^{(j)}(t) = \sum_{n \geq 0} I^G_{a,n} \frac{t^n}{n!}.$$

Thus the full hierarchy is equivalent to coefficientwise nonnegativity of every F^G_a.

Proof. Expand $(X-Y)^{2a}(X+Y)^n$ twice by the binomial theorem and use the independence of X and Y. Expanding only $(X-Y)^{2a}$ after inserting $e^{tX}e^{tY}$ gives the derivative formula. Finally, expansion of $e^{t(X+Y)}$ gives the last equality. The adjoint character is bounded on the compact group, so all series converge absolutely and termwise integration is justified. □

Lemma 1.10 (Central-cover invariance). *If $\pi : \widetilde{G} \to G$ is a finite central covering of compact connected Lie groups, then*

$$I^{\widetilde{G}}_{a,n} = I^G_{a,n} \qquad (a, n \geq 0).$$

Thus every result below depends only on the simple Lie algebra and applies to all of its compact connected global forms.

Proof. The adjoint characters satisfy $\chi^{\widetilde{G}}_{\mathrm{ad}} = \chi^G_{\mathrm{ad}} \circ \pi$, and the push-forward of normalized Haar measure under π is normalized Haar measure. Apply these facts in both variables of the defining integral. □

Lemma 1.11 (Nonnegative cumulants give nonnegative moments). *Let*

$$K(t) = \sum_{r \geq 1} \kappa_r \frac{t^r}{r!} \quad \textit{with} \quad \kappa_r \geq 0,$$

and write

$$e^{K(t)} = \sum_{n \geq 0} q_n \frac{t^n}{n!}.$$

Then $q_n \geq 0$ for every n.

Proof. The exponential formula gives

$$q_n = \sum_{\substack{k_1, \dots, k_n \geq 0 \\ k_1 + 2k_2 + \cdots + nk_n = n}} n! \prod_{r=1}^{n} \frac{1}{k_r!} \left(\frac{\kappa_r}{r!}\right)^{k_r}.$$

Every summand is nonnegative. □

Lemma 1.12 (Adjoint-character normalization). *For G compact and connected with simple Lie algebra,*
$$\int_G \chi_{\mathrm{ad}}(g)\,d\mu_G(g) = 0, \qquad \int_G \chi_{\mathrm{ad}}(g)^2\,d\mu_G(g) = 1, \qquad \max_G \chi_{\mathrm{ad}} = \dim \mathfrak{g}.$$

Proof. The complexified adjoint representation is irreducible: a G-invariant complex subspace differentiates, by connectedness, to an $\mathrm{ad}(\mathfrak{g}_{\mathbb{C}})$-invariant subspace and hence is an ideal of the simple Lie algebra $\mathfrak{g}_{\mathbb{C}}$. It is nontrivial, so Schur orthogonality gives zero for its character mean. The compact real form has an invariant positive-definite inner product, making the adjoint representation self-dual and its character real. Schur orthogonality therefore also gives $\int_G \chi_{\mathrm{ad}}^2 = \int_G |\chi_{\mathrm{ad}}|^2 = 1$. Finally, an orthogonal operator on the d_G-dimensional real adjoint module has trace at most d_G, with equality at the identity. □

Lemma 1.13 (Even-moment lower bound for a positive cap). *Let X be real-valued with $-c \le X \le d$, let $0 < \varepsilon < d - c$, and put $t = d - \varepsilon$. If $t \ge c$, then for every integer $s \ge 1$,*
$$\mathbf{P}\{X > t\} \ge \frac{\mathbf{E}X^{2s} - t^{2s}}{d^{2s} - t^{2s}}.$$
If moreover $\mathbf{E}X^2 = 1$ and $\varepsilon > 1$, then
$$\mathbf{P}\{|X| < \varepsilon\} \ge 1 - \varepsilon^{-2}.$$

Proof. On the complement of $\{X > t\}$ one has $|X| \le t$, while everywhere $|X| \le d$. Thus, with $p = \mathbf{P}\{X > t\}$,
$$\mathbf{E}X^{2s} \le p d^{2s} + (1-p)t^{2s}.$$
Rearrangement proves the first inequality. The second is Markov's inequality applied to X^2: $\mathbf{P}\{|X| \ge \varepsilon\} \le \varepsilon^{-2}\mathbf{E}X^2$. □

Theorem 1.14 (Fixed-group total-degree tail). *Let G be a compact connected Lie group with simple Lie algebra, put*
$$d_G = \dim \mathfrak{g}, \qquad c_G = -\min_{g \in G} \chi_{\mathrm{ad}}(g),$$
and choose
$$0 < \varepsilon < \frac{d_G - 2c_G}{2}.$$
Define the positive Haar masses
$$p_{G,\varepsilon} = \mu_G\{g : \chi_{\mathrm{ad}}(g) > d_G - \varepsilon\}\, \mu_G\{g : |\chi_{\mathrm{ad}}(g)| < \varepsilon\}$$
and put $q_{G,\varepsilon} = d_G - 2\varepsilon$. If $a \ge 1$, n is odd, and
$$2a + n > \frac{\log(1/p_{G,\varepsilon})}{\log(q_{G,\varepsilon}/(2c_G))},$$
then $I^G_{a,n} > 0$.

Proof. First we verify that all constants in the statement exist. Lemma 1.12 shows that the adjoint character has Haar mean zero and second moment one, so it takes both positive and negative values. Since G is connected, its continuous image $\chi_{\mathrm{ad}}(G)$ is an interval; hence that interval contains zero and $c_G > 0$. Garibaldi–Guralnick–Rains [9, Theorem 2.1] prove $\chi_{\mathrm{ad}}(g) \ge -\operatorname{rank}(G)$. If $r = \operatorname{rank}(G)$, then
$$d_G = r + |R| \ge 3r > 2r \ge 2c_G,$$
because a rank-r root system contains the positive and negative simple roots. Thus the interval prescribed for ε is nonempty and $q_{G,\varepsilon} > 2c_G$. Both sets defining $p_{G,\varepsilon}$ are nonempty open sets: the first contains the identity and the second contains a point at which χ_{ad} vanishes. Haar measure has full support on a compact group: finitely many left translates of any nonempty open set cover G, so such a set cannot have Haar measure zero. Hence $p_{G,\varepsilon} > 0$.

Write $S = X + Y$ and $D = X - Y$. On the negative half-plane $S < 0$, the bounds $X, Y \geq -c_G$ imply

$$|S| \leq 2c_G, \qquad |D| \leq 2c_G.$$

Indeed, if $X \geq Y$, then $S < 0$ gives $X < -Y \leq c_G$, whence $0 \leq X - Y < 2c_G$; the other ordering is symmetric. Therefore the absolute value of the entire negative contribution is at most

$$(2c_G)^{2a+n}.$$

On the positive rectangle

$$X > d_G - \varepsilon, \qquad |Y| < \varepsilon,$$

one has $D > q_{G,\varepsilon}$ and $S > q_{G,\varepsilon}$. Its contribution is strictly greater than

$$p_{G,\varepsilon} q_{G,\varepsilon}^{2a+n}.$$

The remainder of the half-plane $S \geq 0$ is nonnegative because n is odd and $2a$ is even. Consequently

$$I_{a,n}^G > (2c_G)^{2a+n} \left[p_{G,\varepsilon} \left(\frac{q_{G,\varepsilon}}{2c_G} \right)^{2a+n} - 1 \right],$$

and the assumed logarithmic inequality makes the bracket positive. □

Corollary 1.15 (Each fixed group has only a finite residual box)**.** *For a fixed compact connected G with simple Lie algebra, only finitely many pairs (a, n) with $a \geq 2$ and odd n remain after combining the two-minus Theorem 1.4 with Theorem 1.14.*

Proof. Theorem 1.4 supplies $a = 1$. The preceding theorem supplies every pair with total degree $2a+n$ above one fixed finite threshold. There are only finitely many nonnegative integer pairs below that threshold. □

2. Complete closure in type A_1

Lemma 2.1 (An effective $SU(2)$ Haar rectangle)**.** *For the adjoint character of* $\mathrm{SU}(2)$ *one may take*

$$c_G = 1, \qquad d_G = 3, \qquad \varepsilon = \frac{1}{8}, \qquad q_{G,\varepsilon} = \frac{11}{4},$$

and the Haar mass in Theorem 1.14 satisfies

$$p_{G,1/8} > \frac{3789107}{38106288000} > \left(\frac{8}{11} \right)^{29}.$$

Consequently $I_{a,n}^{\mathrm{SU}(2)} > 0$ whenever $a \geq 1$, n is odd, and $2a + n \geq 29$.

Proof. Parametrize conjugacy classes by $0 \leq \alpha \leq \pi$. The normalized class density and adjoint character are

$$d\mu(\alpha) = \frac{2}{\pi} \sin^2 \alpha \, d\alpha, \qquad \chi_{\mathrm{ad}}(\alpha) = 1 + 2\cos(2\alpha) = 3 - 4\sin^2 \alpha.$$

Thus $c_G = 1$ and $d_G = 3$. Put

$$A_0 = (0, 1/6) \cup (\pi - 1/6, \pi).$$

Since $\sin \alpha < \alpha$ on $(0, 1/6)$, every point of A_0 satisfies $\chi_{\mathrm{ad}} > 3 - 1/8$. The elementary identity

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} \, dx > 0$$

gives $1/\pi > 7/22$. Taylor's theorem gives $\sin x \geq x - x^3/6$ for $0 \leq x \leq 1/6$, and hence

$$\begin{aligned} \mu(A_0) &= \frac{4}{\pi} \int_0^{1/6} \sin^2 x \, dx \\ &> \frac{14}{11} \int_0^{1/6} (x - x^3/6)^2 \, dx = \frac{541301}{277136640}. \end{aligned}$$

Next put

$$B_0 = \{\alpha : |\alpha - \pi/3| < 1/32\} \cup \{\alpha : |\alpha - 2\pi/3| < 1/32\}.$$

The character vanishes at the two interval centers and $|\chi'_{\mathrm{ad}}(\alpha)| \le 4$, so $|\chi_{\mathrm{ad}}| < 1/8$ on B_0. For $0 \le u \le 1/32$,

$$\sin(\pi/3 - u) = \frac{\sqrt{3}}{2}\cos u - \frac{1}{2}\sin u > \frac{5}{6}\left(1 - \frac{1}{2048}\right) - \frac{1}{64} = \frac{10043}{12288} > \frac{4}{5}.$$

Here we used $\sqrt{3} > 5/3$, $\cos u \ge 1 - u^2/2$, and $\sin u \le u$. Symmetry gives the same bound on the second interval. Their total length is 1/8, so

$$\mu(B_0) > \frac{2}{\pi}\frac{1}{8}\frac{16}{25} > \frac{14}{275}.$$

It follows that

$$p_{G,1/8} \ge \mu(A_0)\mu(B_0) > \frac{3789107}{38106288000}.$$

The second displayed comparison in the statement follows from the exact integer calculation

$$3789107\,11^{29} - 38106288000\,8^{29} = 1140332041100770569516131911193583337 > 0.$$

It is independently checked afresh by `character_ring_iter/verify_full_q3_su2_gmp.cpp`. Since $q_{G,1/8}/(2c_G) = 11/8$, the fixed-group tail theorem applies at every total degree at least 29. □

Proposition 2.2 (Exact $SU(2)$ residual certificate). *For every $a \ge 2$ and every odd $n \ge 1$ with*

$$2a + n \le 27,$$

one has $I^{\mathrm{SU}(2)}_{a,n} > 0$. There are exactly 78 such pairs; their smallest value is

$$I^{\mathrm{SU}(2)}_{2,1} = 16.$$

Proof. Let $c_k(j)$ be the multiplicity of the spin-j irreducible in the kth tensor power of the spin-one representation. The $SU(2)$ Clebsch–Gordan rule gives the exact integer recurrence

$$c_0(j) = [j = 0], \qquad c_{k+1}(0) = c_k(1), \qquad c_{k+1}(j) = c_k(j-1) + c_k(j) + c_k(j+1) \quad (j \ge 1).$$

The adjoint moment is $m_k = c_k(0)$. Substitution in Lemma 1.9 reduces every claimed inequality to a finite integer sum.

The proof companion is

`character_ring_iter/verify_full_q3_su2_gmp.cpp`.

It implements precisely this recurrence and the double sum in Lemma 1.9. It uses GMP integers throughout, checks every generated moment against the independent Catalan binomial transform

$$m_k = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} \frac{1}{j+1}\binom{2j}{j},$$

which follows from $\chi_{\mathrm{ad}} = |\operatorname{tr} U|^2 - 1$, and checks the independently known prefix

$$(m_0, \ldots, m_9) = (1, 0, 1, 1, 3, 6, 15, 36, 91, 232),$$

enumerates the complete quantified domain, asserts that its cardinality is 78, rejects any nonpositive value, and asserts the stated exact minimum. It also cross-multiplies the rational tail comparison in Lemma 2.1. The command `make full-q3-su2-audit` compiles the source with warnings promoted to errors and reruns every check. Its terminal success marker is `SU2_FULL_Q3 VERIFICATION: ALL PASS`. □

Theorem 2.3 (Full adjoint-generated Q_3 in type A_1). *Let G be any compact connected group with Lie algebra of type A_1. Then Ginibre's full condition Q_3 holds on $\mathcal{Q}_G$.*

Proof. By Lemma 1.10, it suffices to use $G = \mathrm{SU}(2)$. The automatic sectors are covered by Theorem 1.3. The row $a = 1$ is Theorem 1.4. For $a \geq 2$ and odd n, the total degree $2a + n$ is odd. Proposition 2.2 covers $2a + n \leq 27$, while Lemma 2.1 covers $2a + n \geq 29$. Hence the complete generator hierarchy is nonnegative, and Theorem 1.3 promotes it to $\mathcal{Q}_G$. □

3. Complete closure in type A_2

Proposition 3.1 (Exact $SU(3)$ cap and residual certificate)**.** *For the adjoint character of* SU(3),

$$m_{54} = 10737197746325886490459958168967522545876 80.$$

This moment gives

$$\mathbf{P}\{X > 6\} \geq \frac{m_{54} - 6^{54}}{8^{54} - 6^{54}}, \qquad \mathbf{P}\{|X| < 2\} \geq \frac{3}{4},$$

and the exact comparison

$$\frac{3}{4}\frac{m_{54} - 6^{54}}{8^{54} - 6^{54}} \cdot 2^{29} > 1$$

holds. Moreover,

$$I^{\mathrm{SU}(3)}_{a,n} > 0 \quad \textit{for all } a \geq 2,\ n \textit{ odd},\ 2a + n \leq 27.$$

There are exactly 78 *inequalities in the last display, and their smallest value is* $I^{\mathrm{SU}(3)}_{2,1} = 72$.

Proof. For Haar $U \in \mathrm{SU}(3)$, put $Z = |\operatorname{tr} U|^2$. Scalar invariance identifies its moments with the corresponding U(3) moments, and Schur–Weyl duality gives

$$E_j := \mathbf{E}Z^j = \sum_{\substack{\lambda \vdash j \\ \ell(\lambda) \leq 3}} (f^\lambda)^2, \qquad m_k = \mathbf{E}(Z-1)^k = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} E_j,$$

where f^λ is computed by the hook-length formula.

The proof companion is

`character_ring_iter/verify_full_q3_su3_gmp.cpp`.

It enumerates every partition in these formulas and performs all arithmetic with GMP integers. It reconstructs $m_0, \ldots, m_{54}$, checks the prefix

$$(m_0, \ldots, m_9) = (1, 0, 1, 2, 8, 32, 145, 702, 3598, 19280),$$

and independently verifies every generated moment against the proved recurrence

$$(k+3)^2 m_{k+1} = k(7k + 11)m_k + 8k(k+1)m_{k-1} \qquad (k \geq 1).$$

It asserts the displayed value of m_{54} and cross-multiplies the tail comparison exactly.

For the residual box it substitutes the reconstructed moments into Lemma 1.9, enumerates the complete 78-pair domain, rejects any nonpositive result, and asserts the exact minimum. The command `make full-q3-su3-audit` rebuilds and reruns the verifier. Its terminal marker is `SU3_FULL_Q3 VERIFICATION: ALL PASS`.

It remains only to justify that the probability bounds consumed by the integer comparison are valid. The identity $\chi_{\mathrm{ad}}(U) = |\operatorname{tr} U|^2 - 1$ gives $-1 \leq X \leq 8$, and equality at -1 occurs at $\operatorname{diag}(1, \omega, \omega^2)$ for a primitive cube root ω. Apply Lemma 1.13 with $(c, d, \varepsilon, s) = (1, 8, 2, 27)$. Its two conclusions are precisely the two displayed probability bounds. □

Theorem 3.2 (Full adjoint-generated Q_3 in type A_2)**.** *Let G be any compact connected group with Lie algebra of type A_2. Then Ginibre's full condition Q_3 holds on $\mathcal{Q}_G$.*

Proof. Central-cover invariance reduces the proof to $G = \mathrm{SU}(3)$. The automatic sectors and cone promotion are given by Theorem 1.3, and the $a = 1$ row is Theorem 1.4. Let $a \geq 2$ and let n be odd. The character bounds are $c_G = 1, d_G = 8$. Taking $\varepsilon = 2$ in Theorem 1.14 gives $q_{G,\varepsilon}/(2c_G) = 2$. The exact probability comparison in Proposition 3.1 therefore proves $I_{a,n} > 0$ for total degree $2a + n \geq 29$. The same proposition supplies every odd total degree at most 27. This exhausts the hierarchy and proves the theorem. ☐

4. Complete closure in types A_3 and A_4

Proposition 4.1 (Exact $SU(4)$ and $SU(5)$ cap and residual certificates)**.** *Put*

$$M_4 = m_{94}^{\mathrm{SU}(4)}, \qquad M_5 = m_{96}^{\mathrm{SU}(5)}.$$

The following exact rational inequalities hold:

$$\frac{15}{16}\frac{M_4 - 11^{94}}{15^{94} - 11^{94}}\left(\frac{7}{2}\right)^{27} > 1,$$

and

$$\frac{285}{289}\frac{2^{96}M_5 - 31^{96}}{48^{96} - 31^{96}}\left(\frac{7}{2}\right)^{37} > 1.$$

In addition,

G	*residual domain*	*number of pairs*	*minimum*
SU(4)	$a \geq 2,\ n$ *odd*, $2a + n \leq 25$	66	$I_{2,1} = 94,$
SU(5)	$a \geq 2,\ n$ *odd*, $2a + n \leq 35$	136	$I_{2,1} = 96.$

Every residual value in the table is strictly positive.

Proof. For $N = 4, 5$, Schur–Weyl duality and the hook-length formula give the exact source

$$E_j^{(N)} = \sum_{\substack{\lambda \vdash j \\ \ell(\lambda) \leq N}} (f^\lambda)^2, \qquad m_k^{(N)} = \sum_{j=0}^{k} (-1)^{k-j}\binom{k}{j} E_j^{(N)}.$$

The proof companion is

`character_ring_iter/verify_full_q3_su45_gmp.cpp`.

It enumerates these partitions, evaluates every hook product and binomial transform with GMP integers, and reconstructs the moments through degrees 94 and 96. As independent checks, it verifies the prefixes

$$(m_0^{(4)}, \ldots, m_9^{(4)}) = (1, 0, 1, 2, 9, 43, 245, 1557, 10829, 80958),$$
$$(m_0^{(5)}, \ldots, m_9^{(5)}) = (1, 0, 1, 2, 9, 44, 264, 1824, 14210, 122224),$$

and every instance in range of the two proved low-rank recurrences from Parts I–II. It then substitutes the moments in Lemma 1.9, checks the complete residual domains and their exact minima, and cross-multiplies both displayed tail inequalities. No floating-point comparison is used. The command `make full-q3-su45-audit` rebuilds the verifier; its terminal marker is `SU45_FULL_Q3 VERIFICATION: ALL PASS`.

To connect the exact comparisons to Haar probabilities, use $\chi_{\mathrm{ad}} = |\operatorname{tr} U|^2 - 1$, so $c_G = 1$ and $d_G = N^2 - 1$. Trace-zero elements exist in both groups. For $SU(4)$ apply Lemma 1.13 with

$$(d, \varepsilon, s) = (15, 4, 47).$$

It gives cap mass at least $(M_4 - 11^{94})/(15^{94} - 11^{94})$ and central mass at least $15/16$; the tail ratio is $(d - 2\varepsilon)/(2c_G) = 7/2$. For $SU(5)$ use

$$(d, \varepsilon, s) = (24, 17/2, 48).$$

After multiplying numerator and denominator by 2^{96}, its cap bound is $(2^{96}M_5 - 31^{96})/(48^{96} - 31^{96})$; the central mass is at least $285/289$, and the tail ratio is again $7/2$. These are exactly the quantities checked by the verifier. $\square$

Theorem 4.2 (Full adjoint-generated Q_3 in types A_3 and A_4)**.** *Let G be a compact connected group whose simple Lie algebra has type A_3 or A_4. Then Ginibre's full condition Q_3 holds on $\mathcal{Q}_G$.*

Proof. Central-cover invariance reduces the two cases to $SU(4)$ and $SU(5)$. The automatic sectors and cone promotion follow from Theorem 1.3; the $a = 1$ row is supplied by Theorem 1.4. For $SU(4)$, the first exact comparison of Proposition 4.1 and Theorem 1.14 supply every odd total degree at least 27, while the residual certificate supplies every odd total degree at most 25. The corresponding split for $SU(5)$ is $2a + n \geq 37$ and $2a + n \leq 35$. Thus no generator case remains in either group, and Theorem 1.3 completes the proof. $\square$

5. Stable type A

Theorem 5.1 (Full stable-rank type-A hierarchy)**.** *Let $G = \mathrm{SU}(N)$ and let $a \geq 1$, $n \geq 0$. If*

$$N \geq 2a + n,$$

then $I^G_{a,n} > 0$. In fact,

$$F^G_a(t) = (2a)!\,\frac{e^{-2t}}{(1-t)^{2a+2}} \quad \textit{through degree } n.$$

Proof. For every $r \leq N$, Schur–Weyl duality gives

$$\mathbf{E}_{\mathrm{SU}(N)}|\operatorname{tr} U|^{2r} = r!.$$

Since $\chi_{\mathrm{ad}}(U) = |\operatorname{tr} U|^2 - 1$, all adjoint moments through degree N agree with those of $A = E - 1$, where E is exponential with mean one. The integral $I_{a,n}$ uses no moment above degree $2a + n$, so the rank hypothesis permits replacement of X, Y by independent copies of A.

Let E_1, E_2 be independent mean-one exponentials, put $R = E_1 + E_2$ and $T = E_1/R$. Then R and T are independent, R has the gamma density $re^{-r}\,dr$, and T is uniform on $[0, 1]$. Moreover,

$$X - Y = R(2T - 1), \qquad X + Y = R - 2.$$

Consequently

$$\begin{aligned} F^G_a(t) &= \mathbf{E}(2T-1)^{2a}\,\mathbf{E}\big[R^{2a}e^{t(R-2)}\big] \\ &= \frac{1}{2a+1}e^{-2t}\frac{(2a+1)!}{(1-t)^{2a+2}} = (2a)!\frac{e^{-2t}}{(1-t)^{2a+2}}. \end{aligned}$$

The logarithm of the last factor after removal of the positive constant is

$$-2t - (2a+2)\log(1-t).$$

Its first cumulant is $2a$, and its rth cumulant for $r \geq 2$ is $(2a+2)(r-1)!$. They are all positive, so Lemma 1.11 proves strict positivity of every coefficient. $\square$

Lemma 5.2 (Paley–Zygmund cap bound)**.** *If $W \geq 0$ has finite second moment and $0 \leq t < \mathbf{E}W$, then*

$$\mathbf{P}\{W > t\} \geq \frac{(\mathbf{E}W - t)^2}{\mathbf{E}W^2}.$$

Proof. Write $A = \{W > t\}$. Then

$$\mathbf{E}W \leq t + \mathbf{E}(W\mathbf{1}_A) \leq t + (\mathbf{E}W^2)^{1/2}\mathbf{P}(A)^{1/2}$$

by Cauchy–Schwarz. Rearrangement and squaring prove the claim. $\square$

Lemma 5.3 (Polynomial one-sided tail bound)**.** *Let X be real-valued, let $T \in \mathbb{R}$, and let P be a real polynomial for which*

$$A = \mathbf{E}\big[(X-T)P(X)^2\big] > 0, \qquad B = \mathbf{E}\big[(X-T)^2P(X)^4\big] > 0.$$

Then

$$\mathbf{P}\{X > T\} \ge \frac{A^2}{B}.$$

Proof. Put

$$R(x) = 1 - \frac{A}{B}(x-T)P(x)^2.$$

If $x \le T$, then $R(x) \ge 1$, so $\mathbf{1}_{\{x\le T\}} \le R(x)^2$. Therefore

$$\begin{aligned}\mathbf{P}\{X \le T\} &\le \mathbf{E}R(X)^2 \\ &= 1 - 2\frac{A}{B}A + \frac{A^2}{B^2}B = 1 - \frac{A^2}{B}.\end{aligned}$$

Taking complements proves the assertion. □

Lemma 5.4 (One-sided cap propagation for every even-minus sector)**.** *Let X, Y be independent copies of a real random variable satisfying $X \ge -C$ and $\mathbf{E}X^2 = 1$, and let T, δ, p_0 satisfy*

$$\delta > 1, \qquad \mathbf{P}\{X > T\} \ge p_0 > 0, \qquad q := T - \delta > 2C.$$

If K is odd and

$$p_0(1-\delta^{-2})q^K > (2C)^K, \tag{2}$$

then

$$\mathbf{E}[(X-Y)^{2a}(X+Y)^n] > 0$$

for every $a \ge 1$, every odd n, and every $2a+n \ge K$.

Proof. Chebyshev's inequality and $\mathbf{E}Y^2 = 1$ give $\mathbf{P}\{|Y| < \delta\} \ge 1 - \delta^{-2}$. On the product of this event with $\{X > T\}$, both $X - Y$ and $X + Y$ exceed q. On the negative half-plane $X + Y < 0$, the lower support bound gives $|X-Y|, |X+Y| \le 2C$, as in the proof of Theorem 1.14. The rest of the positive half-plane contributes nonnegatively. Hence, for $L = 2a + n$,

$$\mathbf{E}[(X-Y)^{2a}(X+Y)^n] > p_0(1-\delta^{-2})q^L - (2C)^L.$$

The right side is positive at $L = K$ by (2); after division by $(2C)^L$ it is strictly increasing in L because $q/(2C) > 1$. □

Proposition 5.5 (Uniform higher-minus tail for $SU(N)$, $N \ge 6$)**.** *Let $a \ge 1$ and let n be odd. Put $L = 2a + n$ and define*

$$K_N = \begin{cases} 27, & N = 6, \\ 29, & N = 7, 8, \\ 31, & N \ge 9. \end{cases}$$

If $N \ge 6$ and $L \ge K_N$, then

$$I^{\mathrm{SU}(N)}_{a,n} > 0.$$

Proof. Put $Z = |\operatorname{tr} U|^2$ and $X = Z - 1$. For

$$E_r^{(N)} = \mathbf{E}Z^r = \sum_{\substack{\lambda\vdash r \\ \ell(\lambda)\le N}} (f^\lambda)^2,$$

apply Lemma 5.2 to $W = Z^{16}$ and $t = (33/5)^{16}$. The exact inequalities verified below include $E_{16}^{(N)} > t$ and give

$$\mathbf{P}\{X > 28/5\} = \mathbf{P}\{Z > 33/5\} \geq \frac{(E_{16}^{(N)} - (33/5)^{16})^2}{E_{32}^{(N)}}.$$

Since $\mathbf{E}X^2 = 1$, Lemma 1.13 also gives

$$\mathbf{P}\{|X| < 11/10\} \geq \frac{21}{121}.$$

On the product of these two events, both $X - Y$ and $X + Y$ exceed

$$\frac{28}{5} - \frac{11}{10} = \frac{9}{2}.$$

On the negative half-plane $X + Y < 0$, the identity $X = Z - 1 \geq -1$ gives $|X - Y|, |X + Y| \leq 2$. Exactly as in Theorem 1.14, it therefore suffices that

$$\frac{21}{121} \frac{(E_{16}^{(N)} - (33/5)^{16})^2}{E_{32}^{(N)}} \left(\frac{9}{4}\right)^L > 1.$$

The exact GMP verifier

`character_ring_iter/verify_full_q3_sun_ge6_gmp.cpp`

clears denominators and checks this inequality at $L = K_N$ for every $6 \leq N \leq 31$. For $N \geq 32$, Schur–Weyl stability gives $E_{16}^{(N)} = 16!$ and $E_{32}^{(N)} = 32!$, so the single checked stable-row comparison at $N = 32$ applies to every larger N. The left side increases with L, because $9/4 > 1$, completing the tail proof. □

Proposition 5.6 (Exact residual type-A certificate for $N \geq 6$)**.** *For every $6 \leq N \leq 28$ and every pair $a \geq 2$, n odd, satisfying*

$$N < 2a + n < K_N,$$

one has $I_{a,n}^{\mathrm{SU}(N)} > 0$. The quantified domain contains 1315 *pairs. Its smallest value is*

$$I_{2,3}^{\mathrm{SU}(6)} = 3550.$$

Proof. The proof companion is

`character_ring_iter/verify_full_q3_sun_ge6_gmp.cpp`.

For each $6 \leq N \leq 28$, it reconstructs $E_0^{(N)}, \ldots, E_{29}^{(N)}$ by enumerating all partitions of bounded length and evaluating the hook-length formula, then forms

$$m_k^{(N)} = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} E_j^{(N)}.$$

It checks $E_j^{(N)} = j!$ independently whenever $j \leq N$, substitutes the moments in Lemma 1.9, enforces the exact stable/tail complement defining the residual domain, checks the per-rank and total scope counts, rejects every nonpositive value, and asserts the displayed minimum.

The same source reconstructs $E_{16}^{(N)}$ and $E_{32}^{(N)}$ for $6 \leq N \leq 31$, uses $16!, 32!$ at the stable endpoint, and checks the tail comparison of Proposition 5.5 after the exact integer clearing

$$21\left(5^{16} E_{16}^{(N)} - 33^{16}\right)^2 9^{K_N} > 121\, 5^{32} E_{32}^{(N)} 4^{K_N}.$$

The command `make full-q3-sun-ge6-audit` rebuilds and reruns the complete calculation. Its terminal marker is `SUN_GE6_FULL_Q3 VERIFICATION: ALL PASS`. □

Theorem 5.7 (Full adjoint-generated Q_3 in every type A)**.** *Let G be a compact connected group whose simple Lie algebra has type A_r for some $r \geq 1$. Then Ginibre's full condition Q_3 holds on $\mathcal{Q}_G$.*

Proof. Types A_1–A_4 are Theorems 2.3, 3.2, and 4.2. Let $G = SU(N)$ with $N \geq 6$; central-cover invariance handles every other global form. The automatic sectors follow from Theorem 1.3, and Theorem 1.4 supplies $a = 1$.

It remains to take $a \geq 2$, n odd, and $L = 2a + n$. If $L \leq N$, Theorem 5.1 applies. If $L \geq K_N$, Proposition 5.5 applies. A pair in neither range can occur only for $6 \leq N \leq 28$, and every such pair is supplied by Proposition 5.6. Indeed, for $N = 29$ there is no odd integer strictly between 29 and 31, and for $N \geq 30$ the stable and tail ranges overlap. Thus the generator hierarchy is complete for every N. Theorem 1.3 promotes it to $\mathcal{Q}_G$. □

6. Complete closure for the exceptional groups

For an exceptional group G, write $m_j = m_j^G$ and set

$$A_G = m_{2k+1} - T m_{2k}, \qquad B_G = m_{4k+2} - 2T m_{4k+1} + T^2 m_{4k}.$$

The following table fixes all parameters used below. The column M is the largest moment consumed by the tail calculation, and K is an odd total degree.

G	c	T	δ	k	M	K
G_2	2	91/10	6/5	9	38	25
F_4	4	211/10	13/10	12	50	37
E_6	3	98/5	7/5	10	42	29
E_7	7	333/10	13/10	17	70	63
E_8	8	461/10	13/10	23	94	71

Table 1. Parameters for the exceptional higher-minus tail certificates.

Proposition 6.1 (Exact exceptional higher-minus tails)**.** *Take the row of Table 1 corresponding to G. If $a \geq 1$, n is odd, and $2a + n \geq K$, then*

$$I_{a,n}^G > 0.$$

Proof. The bounds $\chi_{\mathrm{ad}} \geq -c$ used in the five rows are rigorous: for G_2, F_4, E_7, E_8 they follow from the rank bound of Garibaldi–Guralnick–Rains [9, Theorem 2.1], and for E_6 the sharper value $\min \chi_{\mathrm{ad}} = -3$ is [9, Theorem 2.3 and Table 1].

Apply Lemma 5.3 with $P(x) = x^k$. It gives

$$\mathbf{P}\{X > T\} \geq \frac{A_G^2}{B_G}.$$

By Lemma 1.12 and Markov's inequality,

$$\mathbf{P}\{|Y| < \delta\} \geq 1 - \delta^{-2}.$$

On the product of these two events,

$$X - Y > T - \delta, \qquad X + Y > T - \delta.$$

On $X + Y < 0$, the common lower bound $X, Y \geq -c$ instead gives $|X - Y|, |X + Y| \leq 2c$, exactly as in Theorem 1.14. Thus, with $L = 2a + n$, it suffices that

$$\rho_G R_G^L > 1, \qquad \rho_G = \frac{A_G^2}{B_G}(1 - \delta^{-2}), \qquad R_G = \frac{T - \delta}{2c}. \tag{3}$$

The exact verifier

`character_ring_iter/verify_full_q3_exceptional_gmp.cpp`

checks $A_G, B_G > 0$, $R_G > 1$, and $\rho_G R_G^K > 1$ by GMP rational arithmetic in every row. Hence (3) holds for every $L \geq K$. □

Proposition 6.2 (Exact exceptional residual certificate). *For each row of Table 1, one has*

$$I_{a,n}^G > 0 \quad \textit{whenever} \quad a \geq 2, \quad n \textit{ is odd}, \quad 2a+n < K.$$

The exact scopes and minima are

G	*largest residual degree*	*number of pairs*	$\min I_{a,n}^G$
G_2	23	55	36
F_4	35	136	36
E_6	27	78	38
E_7	61	435	36
E_8	69	561	36

and every displayed minimum occurs at $(a,n) = (2,1)$.

Proof. The proof companion is

`character_ring_iter/verify_full_q3_exceptional_gmp.cpp`.

It reads the exact invariant-tensor moments already regenerated from the Cartan data by the source audit of Parts I–II. For E_8, the independently regenerated values through degree 70 are joined to the transcribed degrees 71–100 from the ancillary tensor-rank table of Bourjaily–Plesser–Vergu [10, ancillary file `adjoint_tensor_ranks.tex`]; the Parts I–II source replay compares that transcription against its own Racah–Speiser reconstruction through $\chi_{\mathrm{ad}}^{\otimes 50}$.

For every pair in the stated boxes, the verifier evaluates the integer formula of Lemma 1.9, rejects a nonpositive value, checks the five scope counts, and asserts the five minima and their coordinates. Independently, it forms A_G, B_G, ρ_G, R_G as reduced GMP rationals, checks the tail inequality at K, and checks that the same chosen certificate does not yet pass at $K-2$. The command `make full-q3-exceptional-audit` rebuilds and reruns the arithmetic audit; its terminal marker is `EXCEPTIONAL_FULL_Q3 VERIFICATION: ALL PASS`. □

Theorem 6.3 (Full adjoint-generated Q_3 for every exceptional type). *Let G be a compact connected group whose simple Lie algebra has type G_2, F_4, E_6, E_7, or E_8. Then Ginibre's full condition Q_3 holds on $\mathcal{Q}_G$.*

Proof. By central-cover invariance, it suffices to use the compact form employed in the moment computation. The automatic sectors follow from Theorem 1.3, and Theorem 1.4 supplies $a = 1$. For $a \geq 2$ and odd n, Proposition 6.1 applies when $2a+n \geq K$, while Proposition 6.2 applies below K. Thus the entire generator hierarchy holds, and Theorem 1.3 promotes it to $\mathcal{Q}_G$. □

7. Stable classical types

Theorem 7.1 (Full stable classical hierarchy). *Let $a \geq 1$ and $n \geq 0$, and put $L = 2a+n$. The inequality $I_{a,n}^G \geq 0$ holds in each of the following ranges:*

G	*stable-rank condition*
$B_\ell = \mathrm{SO}(2\ell+1)$	$L \leq 2\ell+1$,
$C_\ell = \mathrm{Sp}(2\ell)$	$L \leq 2\ell+1$,
$D_\ell = \mathrm{SO}(2\ell)$	$L \leq \ell-1$.

More precisely, throughout these ranges

$$F_a^G(t) = e^{-t+t^2/2} \sum_{j=0}^{a} \frac{\gamma_{a,j}}{(1-t)^{2j+1}} \quad \textit{through degree } n,$$

where

$$\gamma_{a,j} = \binom{2a}{2j}(2a-2j-1)!!\frac{\binom{2j}{j}}{4^j}(2j)! > 0.$$

Here $(-1)!! = 1$.

Proof. The exact stable joint trace laws of Diaconis–Shahshahani [2, Theorems 4 and 6] identify the stable orthogonal and symplectic adjoint variables with the common law

$$X = \frac{Z_1^2 - 1 + \sqrt{2}Z_2}{2},$$

where Z_1, Z_2 are independent standard normal variables. The sign of Z_2 is immaterial.

We first check that the stated ranks supply every moment used by $I_{a,n}$. For type B, the orthogonal Schur integral criterion [11, Section 3, paragraph preceding Lemma 3.1] applied to $s^s_{(1,1)}$ is stable when $s \le 2\ell + 1$. For type C, the symplectic criterion in the same paragraph is stable when $\ell \ge \lfloor s/2 \rfloor$, equivalently $s \le 2\ell + 1$. In type D, the trace degree of χ^s_{ad} is $2s$; the exact orthogonal trace law and the equality of the $\mathrm{SO}(2\ell)$ and $\mathrm{O}(2\ell)$ integrals apply when $2s < 2\ell$, namely $s \le \ell - 1$. Since Lemma 1.9 uses only moments of total degree at most L, the displayed hypotheses give the stable law throughout the required calculation.

Take independent copies

$$X = \frac{Z_1^2 - 1 + \sqrt{2}Z_2}{2}, \qquad Y = \frac{W_1^2 - 1 + \sqrt{2}W_2}{2}.$$

In polar coordinates for (Z_1, W_1), put

$$R = \frac{Z_1^2 + W_1^2}{2}, \qquad C = \frac{Z_1^2 - W_1^2}{Z_1^2 + W_1^2}.$$

Then R is mean-one exponential, $C = \cos(2\Theta)$ with Θ uniform, and R, C are independent. Also

$$A = \frac{Z_2 + W_2}{\sqrt{2}}, \qquad B = \frac{Z_2 - W_2}{\sqrt{2}}$$

are independent standard normals, independent of (R, C), and

$$X + Y = R - 1 + A, \qquad X - Y = RC + B.$$

Only even powers survive in the conditional expansion of $(RC + B)^{2a}$. Using

$$\mathbf{E}B^{2q} = (2q-1)!!, \qquad \mathbf{E}C^{2j} = \frac{\binom{2j}{j}}{4^j}, \qquad \mathbf{E}[R^{2j}e^{tR}] = \frac{(2j)!}{(1-t)^{2j+1}},$$

and $\mathbf{E}e^{tA} = e^{t^2/2}$ gives the asserted formula for F^G_a.

It remains to prove coefficientwise positivity. For each $j \ge 0$, put

$$H_j(t) = \frac{e^{-t+t^2/2}}{(1-t)^{2j+1}}.$$

Its logarithm has cumulants

$$\kappa_1 = 2j, \qquad \kappa_2 = 2j + 2, \qquad \kappa_r = (2j+1)(r-1)! \quad (r \ge 3).$$

They are nonnegative, so every coefficient of H_j is nonnegative by Lemma 1.11. The coefficients $\gamma_{a,j}$ are strictly positive; hence their finite linear combination has nonnegative coefficients as claimed. $\square$

8. A simultaneous nonstable classical tail

For the rest of the classical argument let $X = \chi_{\mathrm{ad}}(U)$ and $Y = \chi_{\mathrm{ad}}(V)$ for independent Haar elements, and put

$$S = X + Y, \qquad D = X - Y.$$

For $G = B_b, C_b, D_b$, use the defining trace T_1 and the signed square trace

$$\tau_G = \begin{cases} \mathrm{Tr}(U^2), & G = B_b, D_b, \\ -\mathrm{Tr}(U^2), & G = C_b. \end{cases}$$

Thus in all three families

$$X = \frac{T_1^2 - \tau_G}{2}. \tag{4}$$

Lemma 8.1 (Exact classical negative endpoints)**.** *For the standard representatives,*

$$C_{B_b} = b, \qquad C_{C_b} = b, \qquad C_{D_b} = \begin{cases} b, & b \text{ even}, \\ b-2, & b \text{ odd}. \end{cases}$$

Proof. Write the defining eigenvalue pairs as z_j, z_j^{-1}, put $x_j = (z_j + z_j^{-1})/2 \in [-1, 1]$, and set $R = \sum_j x_j$ and $Q = \sum_j x_j^2$. Direct substitution in (4) gives

$$\chi_{\mathrm{ad}}^{B_b} = b + 2R + 2(R^2 - Q), \qquad \chi_{\mathrm{ad}}^{D_b} = b + 2(R^2 - Q), \qquad \chi_{\mathrm{ad}}^{C_b} = 2R^2 + 2Q - b.$$

The first two expressions are affine in each variable separately, so their minima occur at vertices of the cube. At a vertex $Q = b$ and $R = b - 2k$. The type-B value is $2R^2 + 2R - b$, whose minimum is $-b$; the type-D value is $2R^2 - b$, whose minimum is $-b$ for even b and $2 - b$ for odd b. The type-C expression is at least $-b$, with equality at $x_1 = \cdots = x_b = 0$. □

Lemma 8.2 (Root-normalized classical Weyl cap)**.** *Let $G = B_b$ or D_b, put $d = \dim G$, and use the positive roots*

G	κ *in* $Q(\theta) = \sum_{\alpha>0} \alpha(\theta)^2 = \kappa\lvert\theta\rvert^2$	$\lvert W\rvert$	$(d_1, \ldots, d_b)$
B_b	$2b-1$	$2^b b!$	$(2, 4, \ldots, 2b)$
D_b	$2b-2$	$2^{b-1} b!$	$(2, 4, \ldots, 2b-2, b)$.

Set $z_{B_b} = 2^{-b}$ and $z_{D_b} = 1$. Suppose that $Q(\theta) \le R$ lies in the standard eigenangle chart and that throughout this ball

$$\prod_{\alpha>0} \left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2} \right)^2 \ge \eta > 0.$$

Then, for the adjoint character X,

$$\mathbf{P}\{X > d - R\} \ge \frac{z_G \eta}{\lvert W\rvert} \frac{\prod_{i=1}^b d_i!}{(2\pi)^{b/2}\Gamma(d/2+1)} \left(\frac{R}{2\kappa} \right)^{d/2}. \tag{5}$$

If $0 < R < 24$, one may take

$$\eta = \left(1 - \frac{R}{24}\right)^2. \tag{6}$$

For D_b, one may instead take, whenever $R < 12\kappa$,

$$\eta = \left(1 - \frac{R}{12\kappa}\right)^{\kappa}. \tag{7}$$

Proof. The character estimate $2(1-\cos u) \le u^2$ gives $d - X \le Q(\theta)$. If $Q(\theta) = R > 0$, at least one root coordinate is nonzero and $2(1-\cos u) < u^2$ for that coordinate; if $Q(\theta) < R$, the conclusion is immediate. Hence $X > d - R$ throughout the cap. Weyl integration and the stated sine-product bound therefore reduce the cap mass to

$$\frac{z_G \eta}{|W|(2\pi)^b} \int_{Q(\theta)\le R} \prod_{\alpha>0} \alpha(\theta)^2 \, d\theta.$$

The factor $z_{B_b} = 2^{-b}$ occurs because the squared-length-two Macdonald–Mehta normalization replaces each short root e_i by $\sqrt{2}e_i$; all type-D roots already have squared length two. The Macdonald–Mehta identity [8, Theorem 3.1(i)], followed by radial integration of the homogeneous discriminant, gives

$$\int_{Q(\theta)\le R} \prod_{\alpha>0} \alpha(\theta)^2 \, d\theta = (2\pi)^{b/2} \frac{\prod_i d_i!}{\Gamma(d/2+1)} \left(\frac{R}{2\kappa}\right)^{d/2},$$

which proves (5).

For $u = \alpha(\theta)$, the elementary bound $\sin(u/2)/(u/2) \ge 1 - u^2/24$ applies. If $R < 24$, all factors on the right are positive, and

$$\prod_{\alpha>0} (1-u_\alpha)^2 \ge \left(1 - \sum_{\alpha>0} u_\alpha\right)^2, \qquad u_\alpha = \frac{\alpha(\theta)^2}{24},$$

proves (6). In type D, every root has squared length two, so $u_\alpha \le R/(12\kappa)$. Concavity of $\log(1-u)$ on the interval from zero to this upper bound, together with $\sum_\alpha u_\alpha = Q(\theta)/24 \le R/24$, gives (7). $\square$

Lemma 8.3 (All-even-minus trace-tail reduction)**.** *Let $C_G = -\min_G \chi_{\mathrm{ad}}$, and choose $c > 2C_G$, $d > 1$, and $q \in \mathbb{R}$. Define*

$$A = \{|T_1| \ge \sqrt{2c+2d+q},\ \tau_G \le q\}, \qquad B = \{|X| < d\}, \qquad p = \mathbf{P}(A)\mathbf{P}(B).$$

If $a \ge 1$, n is odd, and $L = 2a + n$, then

$$I^G_{a,n} \ge 2pc^L - 2\mathbf{E}(\tau_G)_+^L. \tag{8}$$

More generally, if $L_0 \le L$, then

$$I^G_{a,n} \ge 2pc^L - 2(2C_G)^{L-L_0}\mathbf{E}(\tau_G)_+^{L_0}. \tag{9}$$

Consequently, if L_0 is odd and

$$pc^{L_0} > \mathbf{E}(\tau_G)_+^{L_0}, \tag{10}$$

then $I^G_{a,n} > 0$ for every $a \ge 1$, odd n, and $2a + n \ge L_0$.

Proof. On A, equation (4) gives $X \ge c+d$. Hence on the rectangle $A \times B$ both S and D are at least c. On the transposed rectangle $B \times A$, one has $S \ge c$ and $D \le -c$. These rectangles are disjoint because $X \ge c + d > d$ on A, and each has probability p. Their combined contribution is therefore at least $2pc^L$.

The rest of the half-plane $S \ge 0$ contributes nonnegatively. On $S < 0$, split into $X \le Y$ and $Y < X$. The contributions have equal absolute value. On the first piece $X < 0$ and

$$|D| = Y - X \le -2X, \qquad |S| = -X - Y \le -2X.$$

Equation (4) also gives $-2X \le \tau_G$, so $|D|, |S| \le (\tau_G)_+$. This proves (8) after doubling. Independently, $X, Y \ge -C_G$ implies $|D|, |S| \le 2C_G$ on $S < 0$. Retaining L_0 of the L factors under the first bound and the other $L - L_0$ factors under the second proves (9).

Finally, if (10) holds, then $c > 2C_G$ makes the right side of (9) strictly positive for every $L \ge L_0$. Every relevant total degree is odd, which proves the last assertion. $\square$

Lemma 8.4 (Exponential bound for the square-trace moment)**.** *For every integer $L \geq 1$ and every $v > 0$,*

$$\mathbf{E}(\tau_G)_+^L \leq \left(\frac{L}{ev}\right)^L \mathbf{E}e^{v\tau_G}.$$

For the classical square traces and $C_G \geq 296$, one has in particular

$$\mathbf{E}(\tau_G)_+^L \leq 4e^{20}\left(\frac{L}{4e}\right)^L. \tag{11}$$

Proof. For $z \geq 0$, the maximum of $z^L e^{-vz}$ occurs at $z = L/v$ and equals $(L/(ev))^L$. Multiplication by e^{vz} and expectation prove the first display.

For B_b, the power-two identity of Rains [12, Theorem 1.5, equation (23)] gives

$$\tau_G \overset{\mathrm{d}}{=} 1 + 2\operatorname{Re}\operatorname{Tr} W, \qquad W \sim \operatorname{Haar}(\mathrm{U}(b)).$$

The strong Szegő upper bound of Courteaut–Johansson–Lambert [13, Lemma 2.12] therefore gives $\mathbf{E}e^{v\tau_G} \leq e^{v+v^2}$.

For C_b, D_b, the corresponding power-two decomposition [12, Theorem 1.5, equation (22), and equation (36)] writes $\tau_G - 1$ as a sum, up to harmless sign changes, of two independent centered full orthogonal traces. The Fredholm estimate obtained from [13, Proposition 3.3 and equations (3.4)–(3.6)] is

$$\mathbf{E}e^{v\tau_G} \leq \frac{e^{v+v^2}}{(1-\mathfrak{b}_{d_0}(v))(1-\mathfrak{b}_{d_1}(v))}, \qquad \mathfrak{b}_d(v) = \frac{e^{v^2/(d+1)}}{1-v^2/(d+1)^2}\frac{v^d}{d!},$$

provided the companion trace-norm envelope is below one. When $C_G \geq 296$, both component dimensions are at least 11. At $v = 4$, the directed verifier cited below checks

$$\mathfrak{a}_d(v) = \frac{e^{v^2/(d+1)}v^d}{d!(1-v/(d+1))\sqrt{1-v^2/(d+1)^2}}, \qquad \mathfrak{a}_{11}(4) < 1, \qquad \mathfrak{b}_{11}(4) < \frac{1}{2}.$$

Both quantities decrease with $d \geq 11$: after replacing d by $d+1$, the factorial contributes $4/(d+1)$, while the exponential and each remaining rational factor also decrease. Thus each denominator is at least $1/2$. In every family $\mathbf{E}e^{4\tau_G} \leq 4e^{20}$, and substitution in the first display proves (11). ☐

Lemma 8.5 (Source-audited high-rank trace rectangle)**.** *Put*

$$r = \frac{2000001}{1000000}, \qquad \eta = \frac{359}{2000}, \qquad A_0 = \frac{4571}{2000}.$$

For $G = B_b, C_b, D_b$ with $C = C_G \geq 296$, set

$$c = rC, \qquad d = q = \eta C.$$

Then

$$\mathbf{P}\{|T_1| \geq \sqrt{2c+3d},\ \tau_G \leq d\} \geq \frac{1}{2}e^{-A_0 C}, \qquad \mathbf{P}\{|X| < d\} \geq \frac{3}{4}.$$

Proof. The second inequality is Chebyshev's inequality, since $\mathbf{E}X^2 = 1$ and $d > 2$.

The first is the Rains-square trace certificate of Parts I–II, whose inputs are recalled here. The one-trace density estimate [13, Theorem 1.4], the Mills lower bound, and the checked orthogonal parity shifts give

$$\mathbf{P}\{|T_1| \geq \sqrt{(2r+3\eta)C}\} \geq e^{-A_0 C}.$$

Rains' power-two decomposition and the same density estimate give

$$\mathbf{P}\{\tau_G \geq \eta C\} \leq \exp\left(-\frac{(\eta C-1)^2}{4}\right) + \frac{10(\log(2\lfloor C/2\rfloor))^{1/4}}{\sqrt{\Gamma(2\lfloor C/2\rfloor+1)}} \leq \frac{1}{2}e^{-A_0 C}.$$

The last two displayed comparisons, including the unitary projection in type B, are checked with directed rounding for every $296 \le C \le 10000$ and with explicit derivative signs thereafter by `character_ring_iter/trace_square_cutoff_mpfr.cpp`. Subtracting the square-trace exceptional event proves the result. $\square$

Proposition 8.6 (Full higher-minus tail for large classical rank)**.** *Let $G = B_b, C_b, D_b$ and suppose $C_G \ge 296$. Then*

$$I^G_{a,n} > 0 \qquad (a \ge 1,\ n \text{ odd})$$

outside the stable range of Theorem 7.1. Hence the complete generator hierarchy holds in every such row.

Proof. Let L_0 be the first odd total degree after the stable range. The exact negative endpoints give

$$L_0 = \begin{cases} 2b+3 = 2C_G+3, & G = B_b, C_b, \\ b+1 = C_G+1, & G = D_b,\ b \text{ even}, \\ b = C_G+2, & G = D_b,\ b \text{ odd}. \end{cases}$$

In particular $C_G + 1 \le L_0 \le 2C_G + 3$.

Use the parameters of Lemma 8.5. In the notation of Lemma 8.3, its two probability estimates give

$$p \ge \frac{3}{8} e^{-A_0 C}, \qquad C = C_G.$$

Lemma 8.4 shows that the ratio of the two sides in (10) is at least

$$\frac{3}{32} e^{-A_0 C - 20} \left(\frac{4erC}{L_0} \right)^{L_0}.$$

Since the base is greater than one and $C + 1 \le L_0 \le 2C + 3$, the logarithm of this ratio is bounded below by

$$H(C) = \log \frac{3}{32} - A_0 C - 20 + (C+1) \log \left(\frac{4erC}{2C+3} \right). \tag{12}$$

The directed-rounding companion

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character_ring_iter/verify_full_q3_bcd_high_mpfr.cpp
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checks $H(296) > 8$, $H'(296) > 1/10$, and, more importantly for the monotonicity argument,

$$\log \left(\frac{4er \cdot 296}{2 \cdot 296 + 3} \right) - A_0 > \frac{9}{100}.$$

The factor $4erC/(2C+3)$ is increasing, and the remaining term in

$$H'(C) = -A_0 + \log \left(\frac{4erC}{2C+3} \right) + (C+1) \left(\frac{1}{C} - \frac{2}{2C+3} \right)$$

is positive. Consequently the first two terms in $H'(C)$ have sum greater than $9/100$ for every $C \ge 296$, while the last term is positive. Hence $H'(C) > 0$ throughout that range. Thus (10) holds at L_0, and Lemma 8.3 proves every later odd total degree. Theorem 7.1 supplies the prefix.

The same verifier checks the two Fredholm values consumed in Lemma 8.4. The command `make full-q3-bcd-high-audit` rebuilds both directed-rounding programs and requires their terminal zero-failure markers. The source-matched machine-C transcript `certificates/full_q3/fullq3bcdanalytic0003_machine_c.log` authenticates both sources and records zero compile and run statuses; its adjacent SHA-256 file authenticates the transcript. $\square$

Lemma 8.7 (Finite-rank Fredholm MGF bounds)**.** *Let*

$$d_T = \begin{cases} 2b+1, & G = B_b, \\ 2b, & G = C_b, D_b, \end{cases} \qquad \mathfrak{b}_d(v) = \frac{e^{v^2/(d+1)}}{1 - v^2/(d+1)^2} \frac{v^d}{d!},$$

and put

$$\mathfrak{a}_d(v) = \frac{e^{v^2/(d+1)} v^d}{d!(1 - v/(d+1))\sqrt{1 - v^2/(d+1)^2}}, \qquad D_d(v) = 1 - \mathfrak{b}_d(v).$$

If $0 < v < d_T + 1$ *and* $\mathfrak{a}_{d_T}(v) < 1$*, then*

$$e^{v^2/2} D_{d_T}(v) \le \mathbf{E} e^{\pm v T_1} \le \frac{e^{v^2/2}}{D_{d_T}(v)}. \tag{13}$$

For C_b *put*

$$d_0 = 2\left\lceil \frac{b+1}{2} \right\rceil, \qquad d_1 = 2\left\lceil \frac{b}{2} \right\rceil + 1,$$

and for D_b *put*

$$d_0 = 2\left\lceil \frac{b}{2} \right\rceil, \qquad d_1 = 2\left\lceil \frac{b-1}{2} \right\rceil + 1.$$

Whenever the corresponding two trace-norm conditions hold,

$$\mathbf{E} e^{v\tau_G} \le \begin{cases} e^{v+v^2}, & G = B_b, \\ \dfrac{e^{v+v^2}}{D_{d_0}(v) D_{d_1}(v)}, & G = C_b, D_b. \end{cases} \tag{14}$$

Proof. We spell out the Fredholm step because every finite-middle trace estimate uses it. Courteaut–Johansson–Lambert [13, Proposition 3.3] state their Fredholm identity for every complex ξ, not only for real characteristic-function parameters. Substitution of $\xi = \mp iv$ therefore writes the MGF of the random-eigenvalue trace as a Gaussian exponential times

$$\det(I + QK_{\pm v}Q).$$

Here Q deletes the finite prefix determined by the Jacobi ensemble and $K_{\pm v}$ is the corresponding Bessel Hankel operator. We first make the index conversion explicit for all four Jacobi cases. After applying the projection, the Bessel order in entry (r, s), with $r, s \ge 0$, is

kernel	d	Bessel order
K^{--}	$2n$	$2n + r + s$
K^{-+}, K^{+-}	$2n+1$	$2n + 1 + r + s$
K^{++}	$2n+2$	$2n + 2 + r + s.$

In every row, d is the total component dimension. Entry signs do not affect the following absolute-value estimates.

Apply the Bessel estimate used in the proof of [13, Lemma 3.6, equation (3.4)], followed by

$$(d + r + s)! \ge d!(d+1)^{r+s}, \qquad e^{v^2/(d+r+s+1)} \le e^{v^2/(d+1)}.$$

Writing $x = v/(d+1) < 1$, the squared row sum is a geometric series in x^{2s}, and the sum of its square roots is a geometric series in x^r. Hence, in each of the four cases,

$$\|QK_{\pm v}Q\|_1 \le \frac{e^{v^2/(d+1)} v^d}{d!(1-x)\sqrt{1-x^2}} = \mathfrak{a}_d(v).$$

This is the explicit estimate obtained before, and independently of, the source's simpler sufficient restriction $|z| < 2e^{-5/4}n$. Thus the hypothesis $\mathfrak{a}_{d_T}(v) < 1$ justifies the absolutely convergent Plemelj series

$$\log\det(I + QK_{\pm v}Q) = \sum_{j\ge 1} \frac{(-1)^{j+1}}{j} \operatorname{tr}(QK_{\pm v}Q)^j.$$

For a trace of the j-th power, the cyclic index sum uses the same factorial and Bessel bounds. Each cyclic index occurs twice, so the two geometric sums contribute $(1-x^2)^{-1}$. The calculation of [13, Lemma 3.7, equation (3.6)] therefore gives, for all four rows,

$$\left|\operatorname{tr}(QK_{\pm v}Q)^j\right| \le \mathfrak{b}_d(v)^j, \qquad \sum_{j\ge 1} \frac{1}{j}\left|\operatorname{tr}(QK_{\pm v}Q)^j\right| \le -\log\bigl(1-\mathfrak{b}_d(v)\bigr).$$

Notice that $0 \le \mathfrak{b}_d(v) \le \mathfrak{a}_d(v) < 1$: after writing $x = v/(d+1)$, the ratio $\mathfrak{a}_d/\mathfrak{b}_d$ is $\sqrt{(1+x)/(1-x)}$. Exponentiating the last absolute bound therefore gives

$$D_{d_T}(v) \le \det(I + QK_{\pm v}Q) \le D_{d_T}(v)^{-1}.$$

The determinant is positive because the Gaussian prefactor is positive and the left side of the Fredholm identity is an MGF. This proves the determinant sandwich directly from the cited estimates.

It remains to account for the deterministic eigenvalues. The source formula is written for the random eigenvalues only. In type B, their random trace has mean -1; the fixed $+1$ eigenvalue cancels that shift exactly. Equivalently, the source factor $e^{\mp v}$ cancels the restoration factor $e^{\pm v}$ in the two MGFs. In types C, D there is no deterministic shift. The full defining character is therefore centered in every family, in agreement with character orthogonality, and the Gaussian prefactor is $e^{v^2/2}$. This proves (13) for both signs.

For the square trace, Rains' power-two decompositions cited in Lemma 8.4 give either the type-B unitary trace or two independent orthogonal components of dimensions d_0, d_1 displayed in the statement. Apply the preceding upper determinant bound to each orthogonal component and multiply, using independence. This gives the C, D clause of (14); the type-B clause is the unitary strong Szegő bound already proved in that lemma. □

Lemma 8.8 (Exact defining-trace determinant formulas). *For $s \ge 0$, write $I_k = I_k(2s)$ for the modified Bessel function. If T_1 is the full defining trace, then*

$$\mathbf{E}_{B_b} e^{sT_1} = e^{s} \det[I_{|j-k|} - I_{j+k+1}]_{j,k=0}^{b-1}, \tag{15}$$

$$\mathbf{E}_{B_b} e^{-sT_1} = e^{-s} \det[I_{|j-k|} + I_{j+k+1}]_{j,k=0}^{b-1}, \tag{16}$$

$$\mathbf{E}_{C_b} e^{sT_1} = \det[I_{|j-k|} - I_{j+k+2}]_{j,k=0}^{b-1}, \tag{17}$$

$$\mathbf{E}_{D_b} e^{sT_1} = \frac{1}{2} \det[I_{|j-k|} + I_{j+k}]_{j,k=0}^{b-1}. \tag{18}$$

The type-C and type-D trace laws are symmetric.

Proof. Courteaut–Johansson–Lambert [13, Lemma 3.1] give the four Toeplitz–Hankel determinants for the Jacobi parameters in their equation (3.1). Substitution of $\psi(x) = e^{2sx}$ gives Fourier coefficients $I_k(2s)$. Their E_n^{+-} case is $\mathrm{SO}(2b+1)$ after restoring the fixed $+1$ eigenvalue, giving (15); replacing s by $-s$ gives (16). Their E_n^{++} case simultaneously describes $\mathrm{Sp}(2b)$ and the random part of $\mathrm{O}^-(2b+2)$, giving (17). Their E_n^{--} case is $\mathrm{SO}(2b)$; its determinant equals 2 at $s = 0$, which accounts for the factor $1/2$ in (18). Finally, $-I$ belongs to the type-C and even-orthogonal groups, proving symmetry. □

Lemma 8.9 (Finite layered recovery from an exact MGF). *Let Z be bounded, let $M(s) = \mathbf{E}e^{sZ}$, and fix $s > 0$ and a lower endpoint t. Choose $\lambda_- > 0$, increasing upper endpoints $u_0 < \cdots < u_r$,* *and positive $\lambda_0, \ldots, \lambda_r$. Put*

$$L_- = e^{\lambda_- t}\frac{M(s-\lambda_-)}{M(s)}, \qquad U_i = e^{-\lambda_i u_i}\frac{M(s+\lambda_i)}{M(s)}, \qquad F_i = 1 - L_- - U_i, \qquad w_i = e^{-su_i}.$$

If all F_i are positive, then

$$\mathbf{P}\{Z > t\} \geq M(s)\left[w_r F_r + \sum_{i=0}^{r-1}(w_i - w_{i+1})F_i\right]. \tag{19}$$

An endpoint at or above the upper support of Z may be used with $U_i = 0$.

Proof. Under the tilted probability $d\mathbf{P}_s = e^{sZ}M(s)^{-1}d\mathbf{P}$, exponential Markov gives

$$\mathbf{P}_s\{Z \leq t\} \leq L_-, \qquad \mathbf{P}_s\{Z \geq u_i\} \leq U_i.$$

Thus $\mathbf{P}_s\{t < Z < u_i\} \geq F_i$. On the other hand, summation by parts over the layers

$$(t, u_0),\ [u_0, u_1),\ \ldots,\ [u_{r-1}, u_r)$$

gives

$$\mathbf{E}_s[e^{-sZ}\mathbf{1}_{\{t<Z<u_r\}}] \geq w_r\mathbf{P}_s\{t < Z < u_r\} + \sum_{i=0}^{r-1}(w_i - w_{i+1})\mathbf{P}_s\{t < Z < u_i\}.$$

Insert the preceding lower bounds and multiply by $M(s)$. $\square$

Lemma 8.10 (One-step tilted defining-trace lower bound)**.** *Let $t, h > 0$ and put*

$$s = t + h, \qquad \rho(t,h) = 1 - \frac{e^{-h^2/2}}{D(s)}\left(\frac{1}{D(s-h)} + \frac{1}{D(s+h)}\right),$$

where $D(v) = D_{d_T}(v)$. Assume $s - h > 0$, the sandwich (13) at $s-h, s, s+h$, and $\rho(t,h) > 0$. Then

$$\mathbf{P}\{|T_1| \geq t\} \geq 2D(s)\rho(t,h)\exp\left\{\frac{s^2}{2} - s(t+2h)\right\}. \tag{20}$$

Proof. For $\sigma \in \{+1, -1\}$, tilt probability by $M_\sigma(s)^{-1}e^{\sigma s T_1}$. Exponential Markov at parameter h and the two sides of (13) show that the tilted probability of leaving the interval $t < \sigma T_1 < t + 2h$ is at most $1 - \rho(t,h)$. The choice $s = t + h$ centers the Gaussian comparison at the midpoint of this interval; both exit losses are exactly $e^{-h^2/2}$. On the interval the inverse tilt is at least $e^{-s(t+2h)}$. The lower MGF bound at s therefore gives

$$\mathbf{P}\{\sigma T_1 \geq t\} \geq D(s)\rho(t,h)e^{s^2/2 - s(t+2h)}.$$

The positive and negative tails are disjoint. Summing the two bounds proves the display; no symmetry of the trace law is being assumed. $\square$

Proposition 8.11 (Full higher-minus tail for the finite middle ranks)**.** *The complete generator hierarchy holds in the following classical rows:*

$$B_b\ (22 \leq b \leq 295), \qquad C_b\ (29 \leq b \leq 295), \qquad D_b\ (71 \leq b \leq 297).$$

Proof. Let L_0 be the first odd degree after the stable range, as in the proof of Proposition 8.6. In every row put

$$r = \frac{2000001}{1000000}, \qquad c = rC_G, \qquad d = \frac{3}{2}, \qquad t = \sqrt{2c + 2d + q},$$

and use the family-dependent parameters

G	rank range	q	h
B_b	$22 \leq b \leq 295$	$\frac{21}{5}\sqrt{b}$	$\frac{13}{10}$
C_b	$29 \leq b \leq 295$	$4\sqrt{b}$	$\frac{13}{10}$
D_b	$71 \leq b \leq 297$	$\frac{18}{5}\sqrt{b}$	$\frac{6}{5}$.

Let V_G denote the right side of (20). At

$$v_q = \frac{q-1}{2},$$

exponential Markov and (14) give an upper bound U_G for $\mathbf{P}\{\tau_G \geq q\}$. Hence the event A in Lemma 8.3 has probability at least $V_G - U_G$. Chebyshev gives $\mathbf{P}(B) \geq 5/9$. Finally, at

$$v_m = \frac{\sqrt{1+8L_0}-1}{4},$$

Lemmas 8.4 and 8.7 bound $\mathbf{E}(\tau_G)_+^{L_0}$.

The directed-rounding companion

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evaluates precisely these formulas. It checks every trace-norm domain, the positive ρ value for both one-sided tilts, the strict difference $V_G - U_G$, and

$$\frac{5}{9}(V_G - U_G)c^{L_0} > \mathbf{E}(\tau_G)_+^{L_0}$$

in all 768 rows. Its minimum directed logarithmic margin is

$$1.00472681147944218600\ldots > 0 \quad \text{at } D_{71};$$

the minimum log separation between V_G and U_G is $0.16429898256055109\ldots$ at C_{29}, and the largest trace-norm envelope is $0.39958077942628398\ldots < 1$, also at C_{29}. Thus (10) holds at L_0 in every row. Lemma 8.3 supplies the entire tail and Theorem 7.1 supplies the prefix. ☐

Proposition 8.12 (Additional finite classical tails)**.** *For the rows $B_{18}, \ldots, B_{21}$, let the odd tail onsets be*

$$(K_{18}, K_{19}, K_{20}, K_{21}) = (47, 45, 45, 45).$$

For $D_{31}, \ldots, D_{70}$, in increasing rank order, let the onsets be

$$\begin{aligned}&(49, 51, 51, 51, 51, 53, 53, 55, 53, 55, 55, 57, 55, 57, 57, 59, 59, 61, 61, 61,\\ &\ 61, 63, 61, 63, 63, 65, 63, 65, 65, 67, 67, 69, 67, 69, 69, 71, 71, 71, 71, 73).\end{aligned}$$

Then $I^G_{a,n} > 0$ whenever $a \geq 1$, n is odd, and $2a + n \geq K_b$ in the corresponding row.

Proof. Use $c = (2000001/1000000)C_G$ in every row. For B_{18} take

$$d = \frac{13}{10}, \qquad q = \frac{6}{5}b, \qquad h = \frac{5}{4};$$

for B_{19}, B_{20}, B_{21} take

$$d = \frac{3}{2}, \qquad q = b, \qquad h = \frac{6}{5}.$$

For the type-D rows take $d = 3/2$, $h = 6/5$, and

$$q = \begin{cases} 7b/10, & 31 \leq b \leq 38, \\ 3b/5, & 39 \leq b \leq 52, \\ b/2, & 53 \leq b \leq 70. \end{cases}$$

The probability and moment bounds are exactly those in the proof of Proposition 8.11, now evaluated at the displayed onset rather than at the first post-stable degree. The same directed-rounding verifier checks all 44 rows. Its least logarithmic tail margin is

$$0.07529217823349327949\ldots > 0 \quad \text{at } D_{43},\ K_{43} = 55;$$

its least tilted mass parameter is $0.02269453416254314484\ldots > 0$ at B_{19}. Its least logarithmic separation between the defining-trace lower bound and square-trace upper bound, attained at D_{40}, is

$$0.83270005200917997982\ldots > 0.$$

Consequently (10) holds at every displayed onset. Lemma 8.3 propagates each inequality to all later odd total degrees. □

Lemma 8.13 (Exact finite classical adjoint moments)**.** *Write*

$$e_2^j = \sum_{\lambda \vdash 2j} c_j(\lambda) s_\lambda, \qquad c_j(\lambda) \in \mathbb{Z}_{\geq 0},$$

and let s_j be the stable classical adjoint moment. Then

$$m_j^{B_b} = s_j - \sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b+1}} c_j(2\nu).$$

For type C one has

$$m_j^{C_b} = s_j - \sum_{\substack{\nu \vdash j \\ \ell(\nu) > b}} c_j\big((\nu_1, \nu_1, \nu_2, \nu_2, \ldots)'\big).$$

For type D one has, without a restriction on j,

$$m_j^{D_b} = s_j - \sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b}} c_j(2\nu) + \sum_{\substack{\nu \vdash j-b \\ \ell(\nu) \leq 2b}} c_j\big(\lambda(b,\nu)\big),$$

where, after padding ν by zeros to length $2b$,

$$\lambda(b,\nu) = (1 + 2\nu_1, \ldots, 1 + 2\nu_{2b}).$$

An empty partition sum is zero.

Proof. For both orthogonal families the adjoint representation is $\bigwedge^2 V$, so the displayed Schur expansion is the character decomposition of $(\bigwedge^2 V)^{\otimes j}$; iterated vertical-two-strip Pieri proves $c_j(\lambda) \in \mathbb{Z}_{\geq 0}$. The orthogonal Schur integral criterion in Rains [11, Section 3, paragraph preceding Lemma 3.1] says that the $\mathrm{O}(N)$ integral of s_λ is one precisely for even λ of length at most N, and is zero otherwise. For odd $N = 2b+1$, the central element $-I$ identifies the $\mathrm{O}(N)$ and $\mathrm{SO}(N)$ integrals of every power of $\bigwedge^2 V$. Subtracting the even partitions excluded by the length wall gives the type-B formula.

For type C, the adjoint representation is $\mathrm{Sym}^2 V$. The standard involution ω of the symmetric-function ring sends h_2 to e_2 and s_λ to $s_{\lambda'}$. Thus the coefficient of s_λ in h_2^j is $c_j(\lambda')$. Rains' symplectic criterion in the same paragraph says that the integral of s_λ over $\mathrm{Sp}(2b)$ is one precisely when $\ell(\lambda) \leq 2b$ and every row length of λ occurs an even number of times, and is zero otherwise. Such a partition is uniquely $\lambda = (\nu_1, \nu_1, \nu_2, \nu_2, \ldots)$; its length constraint is $\ell(\nu) \leq b$. Subtraction from the unrestricted stable sum proves the type-C formula.

For $\mathrm{SO}(2b)$, averaging a polynomial character over the identity component equals its $\mathrm{O}(2b)$ average plus its determinant-twisted $\mathrm{O}(2b)$ average. The first term admits the even partitions of length at most $2b$. The second admits precisely the partitions of length $2b$ whose rows are all odd: subtracting one full column leaves an even partition, to which Rains' orthogonal criterion applies. Every such determinant-class partition is uniquely $\lambda(b,\nu)$, and its size equation is $2b + 2|\nu| = 2j$. Hence $|\nu| = j - b$, proving the type-D formula. □

Lemma 8.14 (Bounded Littlewood determinants for the finite classical moments)**.** *Work in the completed ring of symmetric functions over $\mathbb{Q}$, put $e_r = h_r = 0$ for $r < 0$, and define*

$$f_r^e = \sum_{k\in\mathbb{Z}} e_k e_{k+r}, \qquad f_r^h = \sum_{k\in\mathbb{Z}} h_k h_{k+r}, \qquad e = \sum_{k\geq 0} e_k, \qquad \bar{e} = \sum_{k\geq 0} (-1)^k e_k.$$

For $b \geq 1$, let

$$\mathcal{B}_b = \frac{1}{2}\left\{ e \det_{1\leq i,j\leq b}(f_{i-j}^e - f_{i+j-1}^e) + \bar{e} \det_{1\leq i,j\leq b}(f_{i-j}^e + f_{i+j-1}^e) \right\},$$
$$\mathcal{C}_b = \det_{1\leq i,j\leq b}(f_{i-j}^h - f_{i+j}^h),$$
$$\mathcal{D}_b = \frac{1}{2} \det_{1\leq i,j\leq b}(f_{i-j}^e + f_{i+j-2}^e).$$

Then, for every $j \geq 0$,

$$m_j^{B_b} = [m_{(2^j)}]\mathcal{B}_b, \qquad m_j^{C_b} = [m_{(2^j)}]\mathcal{C}_b, \qquad m_j^{D_b} = [m_{(2^j)}]\mathcal{D}_b.$$

Here $[m_{(2^j)}]$ denotes the coefficient of the monomial symmetric function $m_{(2^j)}$, and every displayed completed series is interpreted degree by degree.

Proof. Let $c(\lambda)$ and $r(\lambda)$ denote the numbers of odd columns and odd rows of λ. Hall duality gives

$$\langle h_2^j, F\rangle = [m_{(2^j)}]F.$$

Transpose the admitted partitions in the type-B formula of Lemma 8.13. Since the involution ω interchanges e_2 and h_2, this gives

$$m_j^{B_b} = \left\langle h_2^j, \sum_{\substack{\lambda_1\leq 2b+1\\ c(\lambda)=0}} s_\lambda \right\rangle.$$

Huh–Kim–Krattenthaler–Okada [7, Theorem 1.3, equations (1.9)–(1.10)], specialized at $u = 0$, give respectively the sum of the $c(\lambda) = 0$ and $c(\lambda) = 2b + 1$ sectors and their difference. Taking their half-sum gives exactly $\mathcal{B}_b$.

For type C, transposition gives

$$m_j^{C_b} = \left\langle e_2^j, \sum_{\substack{\lambda_1\leq 2b\\ r(\lambda)=0}} s_\lambda \right\rangle.$$

Equation (1.6) of the same source identifies the Schur sum with $\det(f_{i-j}^e - f_{i+j}^e)$. Apply the Hall-isometric involution ω; it sends e_2 to h_2 and f_r^e to f_r^h. Hall duality then gives the formula for $\mathcal{C}_b$.

Finally, the two components in the type-D formula transpose to the sectors

$$\lambda_1 \leq 2b, \qquad c(\lambda) = 0 \text{ or } c(\lambda) = 2b.$$

The second sector is precisely the determinant-twisted component: its $2b$ columns all have odd length. Equation (1.7) of the cited theorem at $u = 0$ is the sum of these two sectors and equals $\mathcal{D}_b$. A final use of Hall duality proves the type-D identity. □

Lemma 8.15 (Two-power coefficient functional)**.** *Let F be homogeneous of symmetric-function degree $2j$. Define the specialization π_2 on power sums by*

$$\pi_2(p_1) = p, \qquad \pi_2(p_2) = q, \qquad \pi_2(p_r) = 0 \quad (r \geq 3).$$

Then

$$[m_{(2^j)}]F = j!\sum_{a=0}^{j}(2(j-a)-1)!![p^{2(j-a)}q^a]\pi_2(F). \tag{21}$$

Under this specialization,

$$\sum_{r\geq 0}\pi_2(e_r)u^r = \exp(pu - qu^2/2), \qquad \sum_{r\geq 0}\pi_2(h_r)u^r = \exp(pu + qu^2/2),$$

and hence, with $e_{-1} = h_{-1} = 0$*,*

$$re_r = pe_{r-1} - qe_{r-2}, \qquad rh_r = ph_{r-1} + qh_{r-2}.$$

In the last display e_r *and* h_r *abbreviate their specialized images* $\pi_2(e_r)$ *and* $\pi_2(h_r)$*.*

Proof. Expand F in the power-sum basis. A power-sum monomial containing p_r with $r \geq 3$ cannot contain the ordinary monomial $x_1^2 \cdots x_j^2$, so it contributes nothing to $[m_{(2^j)}]F$. The remaining power-sum monomials are $p_1^{2(j-a)}p_2^a$. In their expansion, the a ordered p_2 factors must choose distinct variables, while the remaining $2(j-a)$ ordered p_1 factors must hit each of the other variables twice. Thus the coefficient of $x_1^2 \cdots x_j^2$ is

$$\frac{j!}{(j-a)!}\frac{(2(j-a))!}{2^{j-a}} = j!(2(j-a)-1)!!,$$

which proves (21). The two generating series are the standard power-sum formulas for elementary and complete symmetric functions after applying π_2. Differentiating them with respect to u and comparing coefficients gives the recurrences. □

Proposition 8.16 (Exact polynomial and Weyl-cap tails in the residual rows)**.** *For each row in the following table,* $I^G_{a,n} > 0$ *whenever* $a \geq 1$*,* n *is odd, and* $2a + n \geq K$*:*

G	T	δ	k	K	*method*
B_3	227/20	23/20	8	37	*polynomial*
B_4	331/20	5/4	10	41	*polynomial*
B_5	75/4	6/5	10	55	*polynomial*
C_3	57/5	6/5	8	37	*polynomial*
D_4	57/4	23/20	9	47	*polynomial*
D_5	131/10	6/5	7	33	*polynomial*

G	R	δ	q	K
B_2, C_2	–	6/5	39/5	47
B_6	1053/50	3/2	1386/25	51
D_6	371/20	7/5	921/20	49
D_7	18	8/5	357/5	53
D_9	763/10	8/5	751/10	61

In the right-hand table, $d = \dim G$ *and* $q = d - R - \delta$ *except in the* B_2, C_2 *row, where the cap threshold is* $T = 9$ *and* $q = T - \delta$*.*

Proof. For a polynomial row, let

$$A = m_{2k+1} - Tm_{2k}, \qquad B = m_{4k+2} - 2Tm_{4k+1} + T^2m_{4k}.$$

Lemma 5.3, with $P(x) = x^k$, gives $\mathbf{P}\{X > T\} \geq A^2/B$ whenever $A, B > 0$. The exact verifier cited below regenerates all these moments from Lemmas 8.14 and 8.15 and checks, in reduced GMP rational arithmetic,

$$A > 0, \qquad B > 0, \qquad \frac{A^2}{B}(1-\delta^{-2})(T-\delta)^K > (2C_G)^K.$$

Lemma 5.4 proves the six polynomial rows.

For B_2 and C_2, use the type-B_2 positive roots $\theta_1, \theta_2, \theta_1+\theta_2, \theta_1-\theta_2$. On $[0,1/4]^2$, the inequality $\cos u \geq 1-u^2/2$ gives $X > 9$. Moreover, $4\sin^2(u/2) \geq u^2/4$ and $\pi < 4$ give

$$\begin{aligned}\mathbf{P}\{X>9\} &\geq \frac{1}{512}\left(\frac{1}{4}\right)^4 \int_0^{1/4}\int_0^{1/4} \theta_1^2\theta_2^2(\theta_1+\theta_2)^2(\theta_1-\theta_2)^2\,d\theta_1 d\theta_2 \\ &= \frac{1}{512}\left(\frac{1}{4}\right)^{14}\left(\frac{1}{21}-\frac{2}{25}+\frac{1}{21}\right) = \frac{1}{9019431321600}.\end{aligned}$$

The two rows have the same adjoint-character law because their Lie algebras are isomorphic and Lemma 1.10 removes the global-form choice.

For the remaining four cap rows, apply Lemma 8.2. Its exact rational lower bound $\widehat{p}_G$ is obtained as follows:

$$\begin{aligned}\widehat{p}_{B_6} &= \frac{(1-R/24)^2\prod_{i=1}^6(2i)!}{2^6 6!\,2^6(44/7)^3 39!}\left(\frac{R}{22}\right)^{39}, \\ \widehat{p}_{D_6} &= \frac{(1-R/24)^2(2!4!6!8!10!6!)}{2^5 6!\,(44/7)^3 33!}\left(\frac{R}{20}\right)^{33}, \\ \widehat{p}_{D_7} &= \frac{(1-R/24)^2(2!4!6!8!10!12!7!)}{2^6 7!}\frac{4^{46}46!}{12(22/7)^4 92!}\left(\frac{R}{24}\right)^{45}\frac{433}{500}, \\ \widehat{p}_{D_9} &= \frac{(1-R/192)^{16}(2!4!6!8!10!12!14!16!9!)}{2^8 9!}\frac{(7/5)4^{77}77!}{(44/7)^5 154!}\left(\frac{R}{32}\right)^{76}\frac{193}{125}.\end{aligned}$$

Here R has the row-specific value in the table. These are strict lower bounds in (5): besides $\pi < 22/7$, the odd-dimensional rows use

$$\sqrt{2} < \frac{3}{2}, \quad \sqrt{\frac{3}{4}} > \frac{433}{500} \qquad (D_7),$$

and

$$\sqrt{2} < \frac{10}{7}, \quad \sqrt{\frac{763}{320}} > \frac{193}{125} \qquad (D_9),$$

all verified by squaring positive rationals. The sine factors are (6) in the first three rows and (7) in D_9. The same exact ledger checks $R < 9\kappa$ in all four rows; since $\pi > 3$, this places the Euclidean ball strictly inside the standard eigenangle chart.

Finally, the same GMP verifier checks for each cap row

$$\widehat{p}_G(1-\delta^{-2})q^K > (2C_G)^K,$$

using $C_{B_2} = C_{C_2} = 2$, $C_{B_6} = C_{D_6} = 6$, $C_{D_7} = 5$, and $C_{D_9} = 7$. Lemma 5.4 completes the proof. Every rational comparison in this proof is implemented in

`character_ring_iter/verify_full_q3_bcd_bounded_littlewood_gmp.cpp`.

The accepted transcript `certificates/full_q3/fullq3bcd0010_machine_c.log` asserts the exact scope of six polynomial rows and six rational-cap rows, prints a positive reduced rational margin for each, and ends with exit status zero and the terminal marker `FULL_Q3_BCD_BOUNDED_LITTLEWOOD VERIFICATION: ALL PASS`. □

Proposition 8.17 (Directed exact-MGF tails in the residual rows)**.** *For the ranks in the following displays, let K_b be the corresponding entry:*

b	7	8	9	10	11	12	13	14	15	16	17
$K_b(B)$	53	51	53	57	57	61	61	45	45	45	45

b	4	5	6	7	8	9	10	11	12	13	14	15	16
$K_b(C)$	61	55	39	39	49	49	53	55	53	57	59	55	55

b	17	18	19	20	21	22	23	24	25	26	27	28
$K_b(C)$	57	45	45	45	47	47	49	51	53	55	57	59

and

b	8	10	11	12	13	14	15	16	17	18	19	20
$K_b(D)$	57	51	57	45	45	45	45	43	43	43	45	45

b	21	22	23	24	25	26	27	28	29	30
$K_b(D)$	45	45	45	47	45	47	47	49	47	49.

Then, in every displayed row,

$$I^G_{a,n} > 0 \qquad (a \geq 1,\ n \text{ odd},\ 2a+n \geq K_b).$$

Proof. For each row, the certificate chooses exact rational numbers $r, q, \delta > 0$, sets $c = rb$ and

$$t = \sqrt{2c + 2\delta + q},$$

and applies Lemma 8.9 with eight layers to the exact determinants in Lemma 8.8. In type B the positive and negative defining-trace tails are evaluated separately; in types C and D symmetry doubles the positive tail. Let V be the resulting lower bound for $\mathbf{P}\{|T_1| \geq t\}$.

Rains' power-two decompositions [12, Theorem 1.5, equations (22), (23), and (36)] give an upper bound U for $\mathbf{P}\{\tau_G > q\}$ by exponential Markov. In type B the bound is

$$U \leq \exp\{-\lambda(q-1) + \lambda^2\}.$$

In types C, D, the MGF of $\tau_G - 1$ is a product of two exact orthogonal-component MGFs from Lemma 8.8, with the component sizes and signs specified by the cited decomposition. Thus the event A in Lemma 8.3 has probability at least $V - U$. Every row certificate checks $V - U > 0$.

Chebyshev gives $\mathbf{P}\{|X| < \delta\} \geq 1 - \delta^{-2}$. At the displayed onset $K = K_b$, the certificate bounds

$$\mathbf{E}(\tau_G)_+^K \leq \min\left\{ d_T^K, \left(\frac{K}{ev}\right)^K \mathbf{E}e^{v\tau_G} \right\}, \qquad v = \frac{\sqrt{1+8K}-1}{4},$$

where $d_T = 2b+1$ in type B and $d_T = 2b$ in types C, D; the MGF is the same source-audited product just described. It then checks

$$(1-\delta^{-2})(V-U)c^K > \mathbf{E}(\tau_G)_+^K, \qquad c > 2C_G. \tag{22}$$

Lemma 8.3 propagates each row from K to every later odd total degree.

The proof companion is

`character_ring_iter/verify_full_q3_bcd_low_tail_mpfr.cpp`.

It evaluates each Bessel series

$$I_k(2s) = \sum_{m \geq 0} \frac{s^{2m+k}}{m!(m+k)!}$$

through 180 recurrence updates and encloses the remaining positive tail by a geometric series. It evaluates every determinant by interval Cholesky; the matrices are Gram matrices for positive Jacobi weights, and the program rejects a nonpositive pivot. Every rational operation, square root, exponential, logarithm, Bessel bound, and Cholesky operation is rounded outward at 384-bit MPFR precision. The raw certificate prints the full rational schedule for each row before its result and cross-checks all onsets against the independent header `character_ring_iter/full_q3_bcd_remaining_data.hpp`. It verifies all 58 rows; its least directed logarithmic margin in (22) is

$$0.097119405179705447780417 4\ldots > 0 \quad \text{at } C_{15}.$$

Its terminal marker is `FULL_Q3_BCD_LOW_TAIL VERIFICATION: ALL PASS`. The complete transcript is

`certificates/full_q3/fullq3bcdlowtail0001_optimus.log`, authenticated by its adjacent SHA-256 file. □

Proposition 8.18 (Exact bounded-Littlewood residual certificate)**.** *For a row $G = B_b, C_b, D_b$, put*

$$s_G = \begin{cases} 2b+1, & G = B_b, C_b, \\ b-1, & G = D_b. \end{cases}$$

In the rows

$$B_2, \dots, B_{21}, \qquad C_2, \dots, C_{28}, \qquad D_4, \dots, D_{70},$$

let K_G be the tail onset assigned to that row in Proposition 8.16 or Proposition 8.17. Then

$$I^G_{a,n} > 0 \quad \text{whenever} \quad a \geq 2, \quad n \text{ is odd}, \quad s_G < 2a + n < K_G.$$

This domain contains exactly 17,862 *pairs. Its smallest value is*

$$I^{D_5}_{2,1} = 50.$$

Proof. We describe the exact finite calculation and why it evaluates the formulas in the preceding lemmas. After applying π_2, substitute

$$p = t, \qquad q = zt^2.$$

For the homogeneous degree-$2j$ part of any of $\mathcal{B}_b, \mathcal{C}_b, \mathcal{D}_b$, let $P_{G,j}(z)$ be the coefficient of t^{2j}. It has degree at most j, and Lemma 8.15 becomes

$$m^G_j = j! \sum_{a=0}^{j} (2(j-a)-1)!![z^a]P_{G,j}(z). \tag{23}$$

Thus the values at any $j+1$ distinct rational points determine the moment exactly.

The verifier

`character_ring_iter/verify_full_q3_bcd_bounded_littlewood_gmp.cpp`

works simultaneously through $j = 71$. For each of the integer nodes $z = 0, 1, \dots, 71$, it uses the recurrences in Lemma 8.15 to form the specialized e_r, h_r, f^e_r, f^h_r as truncated power series through degree 142, then evaluates every leading determinant in Lemma 8.14. Lagrange interpolation applies the functional in (23). All of this is done in exact prime fields; the primes exceed 2^{30}, and therefore every integer divided by the recurrences and every nonzero node difference is invertible.

The 31 prime-field evaluations are independent OpenMP tasks. Their residues are merged in one fixed descending-prime order by exact GMP CRT, giving a modulus of 961 bits. If $d_G = \dim G$, invariant multiplicity and $|\chi_{\mathrm{ad}}| \leq d_G$ give the elementary reconstruction bound

$$0 \leq m^G_j \leq d^j_G.$$

The program chooses primes until their product is strictly larger than the largest such bound it consumes, and rejects any reconstructed moment outside its own bound. Hence the CRT representative is the unique integer moment, not a probable reconstruction.

The program next regenerates the stable recurrence, checks every stable prefix, checks the $B_2 = C_2$ Lie-isomorphism row, and substitutes the moments into the exact GMP sum of Lemma 1.9. The row ledger is `character_ring_iter/full_q3_bcd_remaining_data.hpp`; it asserts the stable endpoint, tail onset, and moment endpoint for every one of the 114 rows in the statement. The verifier rejects a nonpositive hierarchy value, an incorrect row ledger, or a scope other than 17,862.

The historical machine-C transcript

`certificates/full_q3/fullq3bcd0010_machine_c.log`.

authenticates the first 12,993 cases, while `certificates/full_q3/fullq3bcd0002_machine_c.log` independently authenticates the remaining 4,869 B/D cases by reverse Pieri. The current unified verifier replaces that partition by one determinant run and must be regenerated by the final-source replay. It reports all 17,862 values positive with the stated minimum and ends with `FULL_Q3_BCD_BOUNDED_LITTLEWOOD VERIFICATION: ALL PASS`, and the replay wrapper records exit status zero. As an implementation-independent control for the original low-row subledger, the earlier reverse-Pieri verifier agrees exactly on all 2,845 moments and all 7,484 residual pairs through degree 40. Only 538 required moments remain above that prefix. The independent modular checker described in the code-and-data section evaluates their determinant formulas using ordinary modular arithmetic and Newton forward differences, rather than the promotion verifier's Montgomery arithmetic, Lagrange weights, and GMP reconstruction. It requires agreement modulo 21 distinct primes whose product has 628 bits. This exceeds the independently checked 607-bit maximum of the bounds d_G^j. Both the candidate and the invariant multiplicity lie in $[0, d_G^j]$, so their difference has absolute value smaller than the modulus; the residue agreements therefore prove equality as integers. The checker then recomputes all remaining 5,509 hierarchy values with GMP integers and requires strict positivity. Thus no degree-41–59 reverse-Pieri frontier is needed; the degree-40 prefix remains the structurally independent overlap test. □

Proposition 8.19 (Exact B_{18}–B_{21} and D_{31}–D_{70} residual certificate)**.** *For every B_b with $18 \le b \le 21$ and every D_b with $31 \le b \le 70$,*

$$I^G_{a,n} > 0 \qquad (a \ge 2,\ n \text{ odd}).$$

Thus the complete generator hierarchy holds in all 44 rows.

Proof. Theorem 7.1 supplies the stable prefix, and Proposition 8.12 supplies the tail. The intervening $a \ge 2$, odd-n box is the B_{18}–B_{21} and D_{31}–D_{70} subledger of Proposition 8.18. The case $a = 1$ is Theorem 1.4. These ranges are exhaustive, proving the proposition.

The older reverse-Pieri source `character_ring_iter/verify_full_q3_bd_residual_gmp.cpp` and its authenticated machine-C transcript remain an algorithmically independent control: they check exactly 137 type-B and 4732 type-D residual pairs, 4869 in total, and report minimum

$$I^{D_{31}}_{8,15} = 1272988116891284109857142182380802 > 0.$$

They are no longer a second mandatory supplier. □

Corollary 8.20 (Completed middle orthogonal rows)**.** *The complete generator hierarchy holds for*

$$B_b \quad (b \ge 18), \qquad D_b \quad (b \ge 31).$$

Proof. Combine Propositions 8.6, 8.11, 8.12, and 8.19 with Theorem 7.1. □

Theorem 8.21 (Full hierarchy in types B, C, and D)**.** *Let G be a compact connected group with simple Lie algebra of type B_b $(b \ge 2)$, C_b $(b \ge 2)$, or D_b $(b \ge 4)$. Then*

$$I^G_{a,n} \ge 0 \qquad (a, n \ge 0).$$

Consequently Ginibre's full condition Q_3 holds on $\mathcal{Q}_G$.

Proof. First consider the low rows $B_2, \dots, B_{17}$, $C_2, \dots, C_{28}$, and $D_4, \dots, D_{30}$. The all-plus and even-n sectors are automatic by Theorem 1.3, while Theorem 1.4 supplies $a = 1$. Let $a \ge 2$, let n be odd, and put $L = 2a + n$. Theorem 7.1 applies when $L \le s_G$. Proposition 8.18 applies when $s_G < L < K_G$. At $L \ge K_G$, the row is covered by either Proposition 8.16 or Proposition 8.17. These three ranges are exhaustive.

For B_b with $b \ge 18$ and D_b with $b \ge 31$, use Corollary 8.20. For C_b with $29 \le b \le 295$, use Proposition 8.11; the same proposition covers the still-needed middle ranges $22 \le b \le 295$ in type B and $71 \le b \le 297$ in type D. All remaining higher ranks satisfy the hypothesis of Proposition 8.6.

Thus the generator hierarchy holds in every row. Central-cover invariance gives every compact connected global form, and Theorem 1.3 promotes the generator to $\mathcal{Q}_G$. $\square$

Theorem 8.22 (Full adjoint-generated Ginibre Q_3)**.** *Let G be any compact connected Lie group with simple Lie algebra. For every integer $r \geq 0$, every tuple $f_1, \dots, f_r \in \mathcal{Q}_G$, and every choice of signs $\varepsilon_i \in \{+1, -1\}$ containing an even number of minus signs,*

$$\int_{G\times G} \prod_{i=1}^{r} \big(f_i(g) + \varepsilon_i f_i(h)\big)\, d\mu_G(g) d\mu_G(h) \geq 0.$$

If the number of minus signs is odd, the integral is zero. Hence $\mathcal{Q}_G$ satisfies Ginibre's condition Q_3 with all of its original finite-tuple and sign-pattern quantifiers.

Proof. Classification leaves type A, types B, C, D, and the five exceptional types. Theorem 5.7, Theorem 8.21, and Theorem 6.3, respectively, prove the full generator hierarchy in those cases. Lemma 1.10 removes the choice of compact connected global form. Finally, Theorem 1.3 gives the asserted cone promotion and the vanishing of every odd-minus sector. $\square$

Remark 8.23 (Exact sense in which the result matches Ginibre)**.** Theorem 8.22 matches the full quantifier structure of Ginibre's 1970 condition Q_3 and uses his Proposition 1 multiplicative-cone promotion. It does not replace $\mathcal{Q}_G$ by the cone of all real continuous positive-definite functions. Such a replacement is impossible: Proposition 1.5 gives the exact SO(3) value $-8/279$.

Code and data availability

Certificate semantics and trusted base. The computer-assisted results have three separate layers. First, each numbered proposition states the mathematical input formula, the finite quantified domain, and the sign or interval comparison to be decided. Second, the named verifier reconstructs the inputs and evaluates that comparison; a stored output value is not accepted as an input merely because it appears in a transcript. Third, the transcript records the source identities, scope counts, arithmetic outputs, terminal marker, and process status of one execution. A transcript authenticates a completed run but is not, by itself, the proof of the formula implemented by the verifier.

The trusted computational base is consequently explicit: a conforming C++ implementation, GMP integer and rational arithmetic, MPFR directed rounding, and the short orchestration and manifest parsers named below. No claim is made that these programs are formally verified. The exact stages avoid probabilistic reconstruction: when prime fields and CRT are used, the proved modulus bound makes the reconstructed integer unique. The interval stages accept a sign only when the outward-rounded lower endpoint is positive. Independent recurrence checks, alternative low-degree implementations, sanitizer runs, and mutation tests are redundancy controls; their stated cutoffs are not represented as full second proofs of computations beyond those cutoffs.

The audited computation package for Parts I–III is rooted at the immutable publication import commit

https://github.com/skypher/ginibre-q3-adjoint/tree/91363c3cebb37a44349f613ab0a5aa6dcd412af3/ginibre_q3.

Its full identifier is

`91363c3cebb37a44349f613ab0a5aa6dcd412af3`.

The immutable import commit controls the execution snapshot that produced the expensive arithmetic transcripts. The historical tag `v1.0.0` contains that import binding, but it is not asserted to contain later manuscript revisions. For the present submission, the canonical source bytes are the files in the submitted archive whose hashes are listed in `certificates/full_q3/full_q3_source_manifest.sha256`. A public release of the final submission must reproduce that manifest exactly.

The formal submission has three numbered parts and a formal detailed supplement, supplied as three inseparable manuscript files: `paper.pdf`, containing the compact Parts I–II, `paper_full.pdf`, containing their load-bearing correction-prefix derivations, and `full_q3_extension.pdf`, containing the present Part III. The archived computation-bearing sources are `paper.tex`, `paper_full.tex`, and `full_q3_extension.tex` in that commit. The immutable import commit authenticates the arithmetic snapshot. The current publication sources are bound directly by `certificates/full_q3/full_q3_source_manifest.sha256` and are the inputs rebuilt by the commands below. The historical aggregate transcript predates publication-only prose and metadata revisions; the fail-closed `verify_full_q3_related_work_rebind.py` records that boundary and rejects any change to an arithmetic-bearing source. That rebind is a provenance test, not a step required to construct the submitted source: the final files are bound directly by the final-source manifest. A final release archive must preserve both the immutable execution snapshot and those exact final-source bytes. Part III is not offered as a standalone proof: its Theorem 1.4 imports Theorem 2.2 of Parts I–II, which is included in the submission and archive.

The detailed computational supplement `paper_full.pdf` is rebuilt from the same source. Its numbered correction-prefix derivations and half-stable bridge reduction are incoming proof nodes for the residual B/C closure; diagnostic and superseded material is excluded from that PDF and remains available only in the source development archive. The compact contract, accepted-certificate boundary, and family closures appear in `paper.pdf`.

The exact replay contract, software requirements, host/thread guidance, and terminal success markers are documented in `REPLAY.md` and `FULL_Q3_DISTRIBUTED_REPLAY.md`. The principal requirements are a C++20 compiler with OpenMP, GMP, MPFR, Python 3, GNU Make, and a LaTeX installation. From the repository root, the three aggregate commands are

```
python3 ginibre_q3/run_publication_short_audits.py
make -C ginibre_q3 clean-room-replay
make -C ginibre_q3 full-q3-extension.
```

The exact validated Ubuntu, compiler, GMP/MPFR, OpenMP, Python, and TeX versions used for the July 2026 publication rebuild are recorded in `ENVIRONMENT.md`; the Makefiles remain authoritative for compilation and link flags. These commands implement three reproducibility tiers. The first command runs the document, dependency, interface, scope, moment-formula, cone-promotion, $SO(3)$, and frontier-formula checks as one fail-closed short suite. The Part III command regenerates its exact and directed-rounding arithmetic; the archived two-worker run took 44 minutes 55 seconds. The separate Parts I–II clean-room command is the full certificate-regeneration tier: its partitioned stages have two- or eight-hour guards and require the machine-C resource envelope documented in `REPLAY.md`. Part III does not silently substitute its faster audit for that imported computation.

For the final public archive, the clean-worktree driver `run_final_publication_replay.sh` runs the preflight and both arithmetic aggregates, records the exact commit and final-source-manifest identity in one persistent log, and rejects any replay that changes the source tree. The deterministic packager `build_final_release_archive.py` then includes every tracked publication file, that successful replay log, and the exact commit metadata in one checksum-addressed archive. The commands and terminal markers are given in `REPLAY.md`. The archive deposited for publication must be the output of this final-source workflow; the historical execution snapshot is retained as audit history, not substituted for the final replay.

The full-range independent check is supplied without constructing the large degree-41–59 reverse-Pieri frontier. Its source is

```
character_ring_iter/verify_full_q3_bcd_modular_moment_checker.cpp.
```

It checks the archived degree-40 reverse-Pieri overlap, verifies the 538 higher candidates by the uniquely bounded 21-prime calculation described above, and checks the 5,509 higher hierarchy values exactly. It rejects any incorrect row, degree, count, residue, character bound, modulus bound, or nonpositive hierarchy value before emitting `FULL_Q3_MODULAR_CHECKER VERIFICATION: ALL PASS`. The optional target `full-q3-extension-independent-controls` runs this checker together with the older reverse-Pieri supplier for the same bounded-Littlewood box. They are independent cross-checks, not additional proof obligations of `full-q3-extension`; `run_full_q3_bcd_independent_audit.sh` is the persistent-log wrapper.

SHA-256 manifests authenticate every load-bearing source, accepted transcript, generated ledger, and cited source artifact. Compiler and resource checks and author self-audits are integrity or redundancy tests; they are not independent peer review and do not by themselves decide a sign. The mathematical certificate claims are the exact integer/rational or directed-rounding comparisons described in the numbered results above.

The executable proof obligations are the following. This list fixes the input/output contract independently of program names.

(1) An *exact moment supplier* receives a root datum or one of the bounded Littlewood determinants in Lemmas 8.13– 8.15; it must output every requested integer moment together with its rank and degree. A CRT output is accepted only after the product modulus exceeds the displayed character bound d_G^j, making the representative unique.
(2) An *exact hierarchy evaluator* receives only those moments and a finite list of triples (G, a, n); it evaluates the double sum of Lemma 1.9, rejects a missing or duplicate triple, and accepts only a strictly positive integer for every listed residual case. The row ledger must prove that the listed triples are exactly the complement of the stable and analytic-tail ranges.
(3) A *directed analytic supplier* receives rational tail parameters and the exact determinant or Fredholm formulas proved above. Every elementary operation, Bessel remainder, determinant pivot, exponential, and logarithm is enclosed with outward MPFR rounding. It accepts only when the final lower endpoint is positive and when all trace-norm and support hypotheses are also certified by intervals.
(4) An *independent control* may consume candidate moments but not a candidate sign. It must reconstruct the same mathematical object by a different recurrence, interpolation, or modular path and then reevaluate the hierarchy formula. Shared formulas and shared input ledgers are part of the trusted base and are not described as independent.

Table 2 gives the compact trusted-computation map. Here "exact" means integer or rational arithmetic, including prime-field evaluation followed by uniquely bounded GMP reconstruction; "directed" means outward-rounded 384-bit MPFR interval arithmetic.

Result block	Program/transcript identifier	Arithmetic	Verified scope
Imported two-minus theorem	`distributed0001`	exact/directed	200 replay stages; all compact connected simple types; every $n \geq 0$
Low and stable type A	`su2`, `su3`, `su45`, `sun-ge6`	exact GMP	all residual boxes and exact tail comparisons; 1,673 residual pairs
Exceptional types	`exceptional`	exact GMP	five tails and 1,265 residual pairs
High and middle classical tails	`bcdanalytic0003`	directed MPFR	all $C_G \geq 296$ rows, 768 middle rows, and 44 supplemental rows
Remaining classical tails	`bcdlowtail0001`	directed MPFR	58 finite-rank tail rows
Finite classical residual boxes	final-source bounded replay; historical `bcd0010`/`bcd0002`	exact prime/GMP	114 rows and 17,862 residual inequalities

TABLE 2. Map from the computer-assisted proof blocks to their accepted program/transcript families and arithmetic boundaries.

The identifiers in the table resolve to the following authenticated transcripts:

- `certificates/full_q3/distributed0001/fullq3distributed0001.log`;
- `certificates/full_q3/fullq3complete0002_optimus.log`;
- `certificates/full_q3/fullq3bcdanalytic0003_machine_c.log`;

- `certificates/full_q3/fullq3bcdlowtail0001_optimus.log`;
- `certificates/full_q3/fullq3bcd0010_machine_c.log`; and
- `certificates/full_q3/fullq3bcd0002_machine_c.log`.

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